

Increasing spatial and temporal resolution in energy system optimisation model – The case of Kenya

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ABSTRACT

At the time of writing, 759 million people (2019) still lack access to electricity globally. It is important for energy planning to describe plausible pathways to achieve national goals, using tools such as energy systems models to explore scenarios and provide insight. Until recently, modelling energy access in countries with a low electrification rate was conducted at low spatial (e.g., national) and/or temporal resolutions (e.g., annual time slices or 'overnight' electrification). In this paper, we develop methods in an open-source computational workflow with high spatial resolution in an open-source energy systems optimisation model. We use Kenya as our case application where approx. 16 million people still lack access to electricity (2019). One reference scenario and two diagnostic hypothetical scenarios are developed to assess the model. The spatial resolution of approximately 40 by 40 km cells leads to 591 demand cells split between electrified and un-electrified population. The results show that in the reference scenario, the optimal supply option for the unelectrified population is PV panels and batteries. At the same, an oversupply of the planned power plants is observed. The model can capture dynamics between spatially explicit supply options and central power plants in one model.

1. Introduction

Energy is an enabler of economic and social development. With the launch of the 17 sustainable development goals (SDGs), clear targets have been set for Agenda 2030 [1]. SDG7, in particular universal access to modern energy, has been identified to be an enabler for education (SDG4), ending poverty (SDG1), healthcare (SDG3), and water and sanitation (SDG6) [2]. The target is still far from being met. In 2019, 759 million people still lacked access to electricity in the world with the majority living in Sub-Saharan Africa [3].

Energy planning is central to achieving energy access. To support planning, energy system optimisation models (ESOMs) are important tools to explore different scenarios and give insight [4,5]. To achieve SDG7 by 2030, decentralised energy supply options have been identified as key, due to the abundant solar potential, decreasing costs for decentralised technologies, and the given timeframe [3]. Energy system modelling frameworks, such as TIMES from IEA-ETSAP are often applied with the assumption that the countries investigated already have a large grid infrastructure, and typically simplify spatial constraints associated with those that are unelectrified [6]. For countries with low electrification rates, grid roll-out is often difficult and spatial characteristics are important for electrification costs and technology choice. With the

energy transition to larger quantities of variable renewable energies and larger shares of distributed technologies, new challenges in energy modelling arise. Most decentralised supply options, such as solar and wind, are intermittent technologies which require a higher temporal resolution to understand the implications of the techno-economic analyses.

The evolution and timing of demand play an important role when designing the energy system. Electricity access is, however, not binary, and socio-economic factors interact with demand to form a complex system [7]. For example, different tiers, or levels of electricity demand generate different load profiles [8]. Several studies have looked at the isolated electricity system under different load profiles to understand the implications of different loads [9–11].

At the same time, increasing decentralised supply options with diverse spatially distributed energy resources requires analysis with an increased spatial resolution compared to traditional models. A review of ESOM by Ref. [12] highlights the need for a higher spatiotemporal granularity in ESOM to better represent variable renewable energy resources. Furthermore, the authors highlight the need for a better representation of specific features in developing countries to allow a specific representation of local challenges [12].

Increasing the spatiotemporal resolution of energy modelling could

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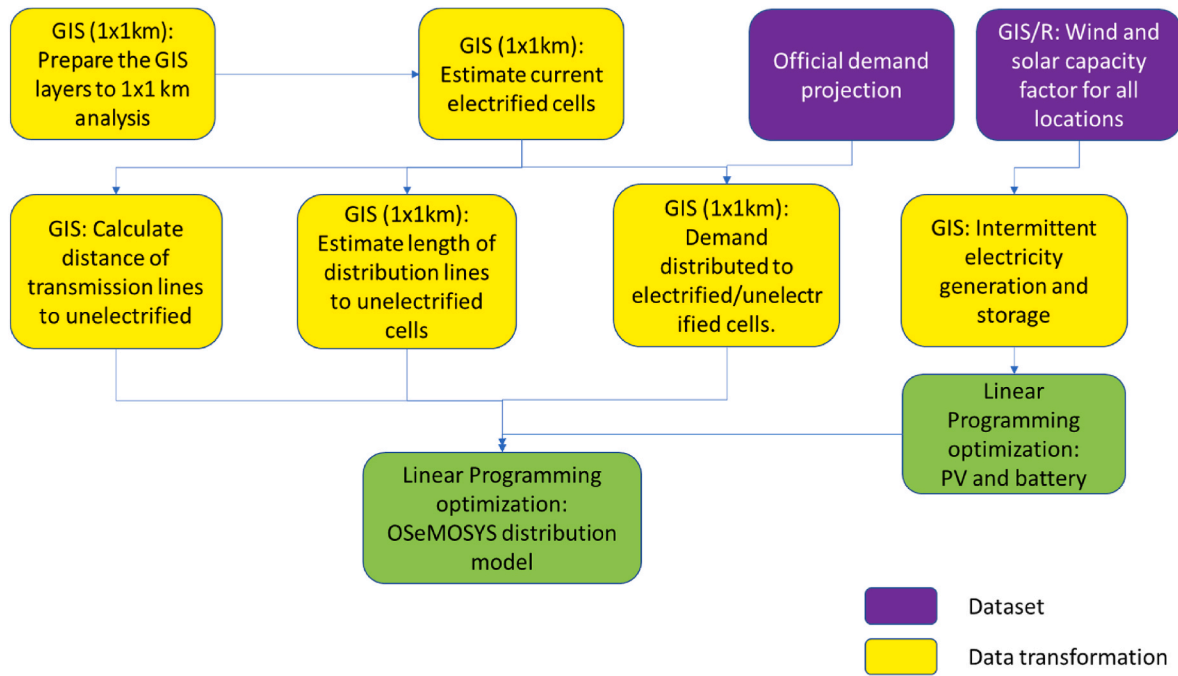


Fig. 1. Modelling workflow.

Table 1
GIS datasets in case application for Kenya.

GIS datasets	Source
Night-time lights	[36]
Population	[37]
Transmission network	[38]
Roads	[39]
Mini-grid	[40]
Substations and transformers	[41,42]
GDP PPP	[43]
Capacity factor wind	[44]
Capacity factor PV	[45]

allow investigation of technology costs, timing, and generation mix across three key dimensions. Socio-economic aspects, such as population, demographics, and income influence the spatial pattern of demand; supply-side aspects, including the changing spatial pattern of renewable resources and location of transmission infrastructure; demand-side aspects to identify local demand variations and regional consumer variations. Increasing the spatial resolution in ESOM often comes with a computational cost which can be motivated in a heterogenous system with spatial variation [4]. Ref. [13] performed a global sensitivity analysis using a Mixed Integer Linear Programming¹ (MILP) heat decarbonisation model. The spatial dimension was one of the uncertain variables together with demand, discount rate, costs, efficiency, network costs and fuel prices. The study found that the spatial resolution was comparable in importance to demand levels and discount rates for the network expansion.

Several models have been developed using a Geospatial Information System (GIS) to address rural electrification planning in terms of the network, power generation sizing and clustering of demand cells [15]. As an example, Network Planner [16] and Open Source Spatial electrification Tool (OnSSET) [17,18] use different heuristic methods

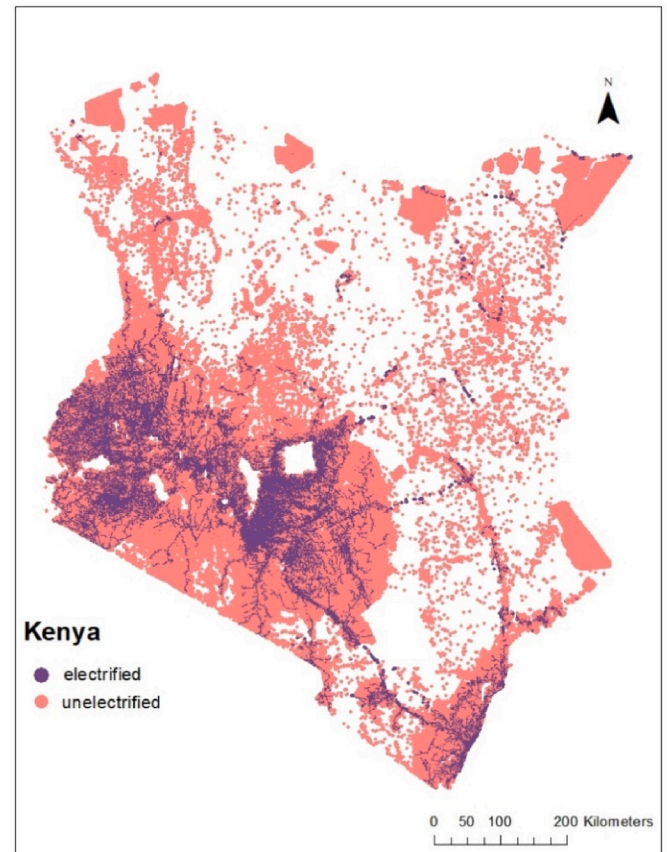


Fig. 2. Electrified and un-electrified settlements, along with regions with no settlements (in white), for Kenya using night-time light, distance to road and electricity network.

¹ A mixed integer linear program is when some of the variables are restricted to be integers while other variables remain continuous. This is commonly used when fractional variables are not realistic [14].

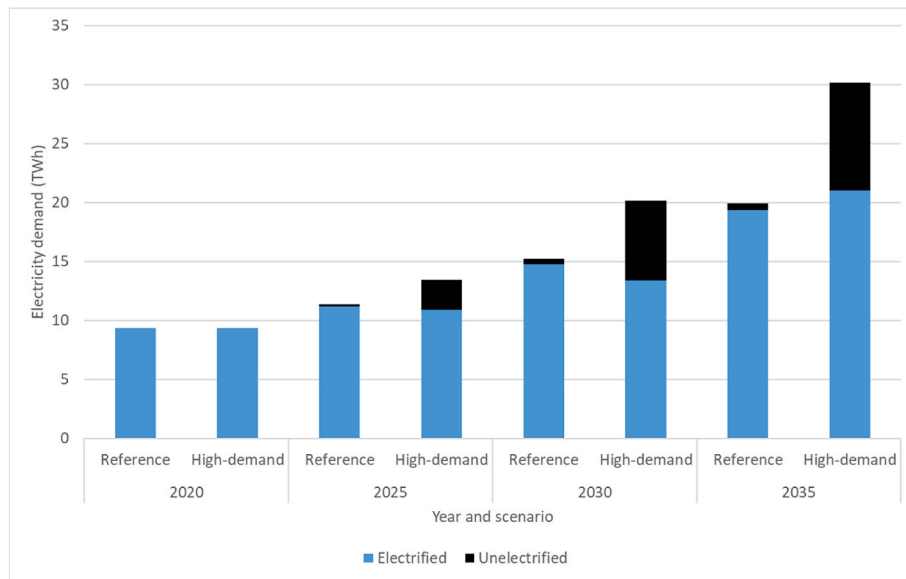


Fig. 3. Modelled demand for reference and high-demand scenario, where the demand for electrified means that those cells had electricity in the base year as opposed to the unelectrified cells that are electrified within the modelling period.

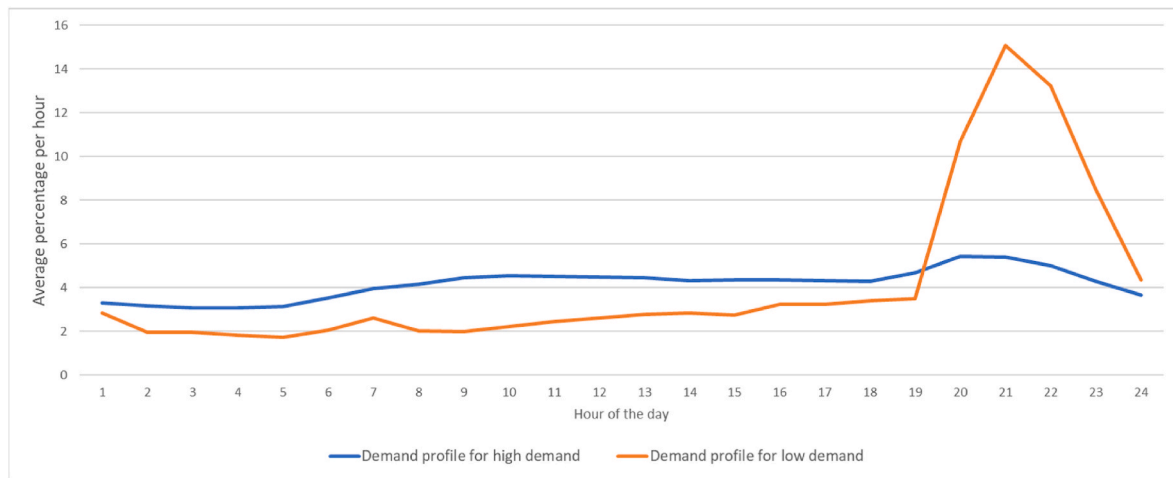


Fig. 4. The average hourly daily demand percentage for one day. The low demand profile is adapted from Ref. [61] and the high demand profile [62].

MILP model for Kenya. It optimises the cost of either extending the grid to an unelectrified cell or supplying off-grid PV technology instead. However, these models treat the grid as a black box, e.g., the power plants supplying the grid are not modelled. A noteworthy initial effort on detailing many loads and decentralised supply options in an ESOM limited itself to the study of a single village [20].

Some studies have linked GIS tools to long-term planning tools (Open Source Energy Modelling System (OSeMOSYS)) which feeds back the grid cost [21,22]. A study by Ref. [23] investigates low carbon futures in Kenya for 47 load centres with a MILP model. However, they do not explicitly model unserved populations nor include off-grid supply options. Demand-side geospatial modelling is also explored by Ref. [24] with an OSeMOSYS model for Tanzania with detailed demand projections following the Multi-Tier framework (MTF) for households.² The resolution for the GIS analysis is 10 by 10 km in the pre-processing step but then aggregated to 6 demands (four rural, one urban and, one

industrial) in the optimisation run. All these models account for the grid in a detailed way, however, the spatial dimension either uses approximate methods (OnSSET) or is modelled on an aggregated level.

For Uganda, Ref. [25] developed a multi-objective MILP model for Uganda for 112 different regions where the transmission and distribution network are integrated into the model, allowing on-grid and off-grid electrification at different percentages. The work by Trotter et al. covers both the grid and a relatively high spatial resolution. The study does not account for variations within a year but instead uses annual values in a multiyear approach.

In this paper, we approach rural electrification planning with long-term energy planning using linear programming (LP). LP computes a globally optimal solution and allows easy exploration of trade-offs between supply and demand as well as marginal costs. A large proportion of ESOMs adapt LP frameworks for this reason [12]. Existing ESOMs such as PyPSA [26] and Calliope [27] account for spatial detail and high temporal resolution, but for a single year [28].

We focus on the problem related to the national planning of rural electrification planning (SDG 7.1, universal access to electricity) in the case of Kenya [29]. In Kenya's Energy Act 2019, all 47 counties are

² International Energy Agency and World Bank, Sustainable Energy for All 2013–2014: Global Tracking Framework Report, 2014.

Table 2
Scenario description.

Scenario	Description of the scenarios used in the modelling
Reference	The demand follows the reference demand from LCPDP 2020–2040. The unelectrified year/rural demand is estimated in LCPDP to be 78 kWh/connection per year with an annual growth of 4 %. The household size is 4 people. With these low demands, the annual rural demand amounts to 3 % of 2030 of the total demand [57,58]. Universal access to electricity is achieved in 2030. Renewable technology costs are expected to decrease over time due to learning curves. The costs follow [65] projections for moderate development.
High-demand	The overall demand follows the Vision 2030 demand from LCPDP 2020–2040. However, the unelectrified demand is modelled at 1000 kWh/connection and year with 6 % annual growth. This scenario serves as a diagnostic scenario to understand how the model results respond to a large demand change as this strongly influences the results. The results also provide insights into the implications to countries where a demand of 1000 kWh is a more reasonable assumption. The household size is 4 people per household. With the demand of 1000 kWh/connection, the rural demand amounts to 34 % of the total demand in 2030. Universal access to electricity is achieved in 2030. As the demand is higher for this scenario the unelectrified demand profile follows the high-demand profile as more appliances are used per household. Renewable technology costs are expected to decrease over time due to learning curves. The projections follow [65] projections for advanced development.
Dry-hydro	This scenario follows the same scenario settings as the High-demand scenario; however, all hydroelectric power is expected to decrease in production due to climate change. This is modelled from 2024 and onwards with reductions which range from –33 % to –37 % depending on the powerplant [57,58].

required to develop county energy plans to define the local energy requirements. At the same time, there is a concern from Kenya Power and Lighting Company (KPLC) that the planned generation investments will lead to a power surplus [30].

Given the importance of the spatial resolution, and the lack of appropriate tools with a multi-year approach and time slices representation for electricity access, and the need for modelling both supply, demand, and network expansion in one model we ask the following questions:

- What is the cost-optimal techno-economic solution for Kenya when representing both spatially explicit decentralised supply and demand and centralised power supply in one ESOM?
- To what degree can an ESOM be extended to represent spatial information and what are the insights and trade-offs from this approach?

The paper is structured as follows, Methods in Section 2, Results in Section 3, Discussion in Section 4, and Conclusions in Section 5.

2. Methods

2.1. Reproducible and FAIR data

The work has for transparency, and reproducibility been developed with an open-source Python and GNU MathProg codebase, with open-access data where the workflow is reproducible from GitHub and where needed, datasets are uploaded to Zenodo³ and follow the FAIR (Findable, Accessible, Interoperable and Reusable) principles [31]. The

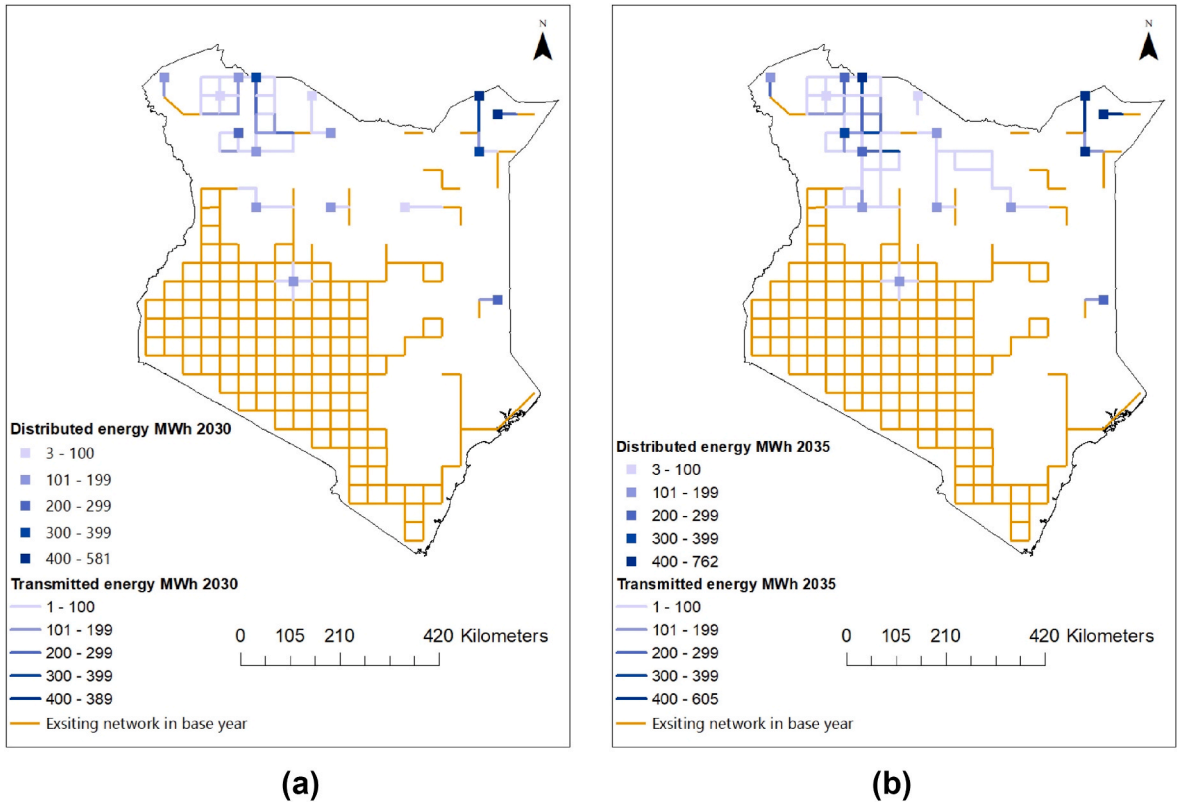


Fig. 5. a, b. Electricity flow between new transmission lines (blue lines), existing transmission lines (yellow lines), and the new distribution lines (squares) in reference scenario over the years 2030 (a) and 2035 (b) (MWh).

³ https://github.com/KTH-dESA/GEOSeMOSYS_Kenya/blob/master/src/input_data/zenodo.txt.

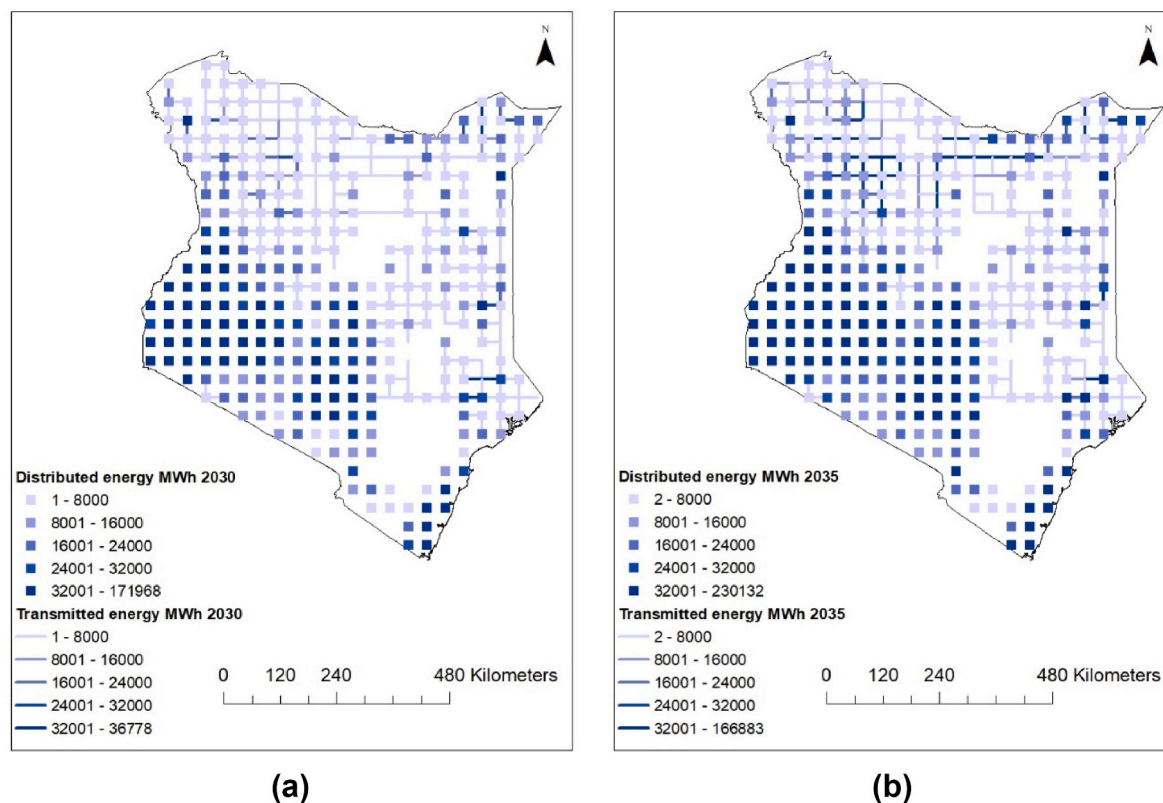


Fig. 6. a, b. Electricity flow between new transmission lines (blue lines) existing transmission lines (yellow lines), and the new distribution lines (squares) in high demand scenario over the years 2030 (a) and 2035 (b).

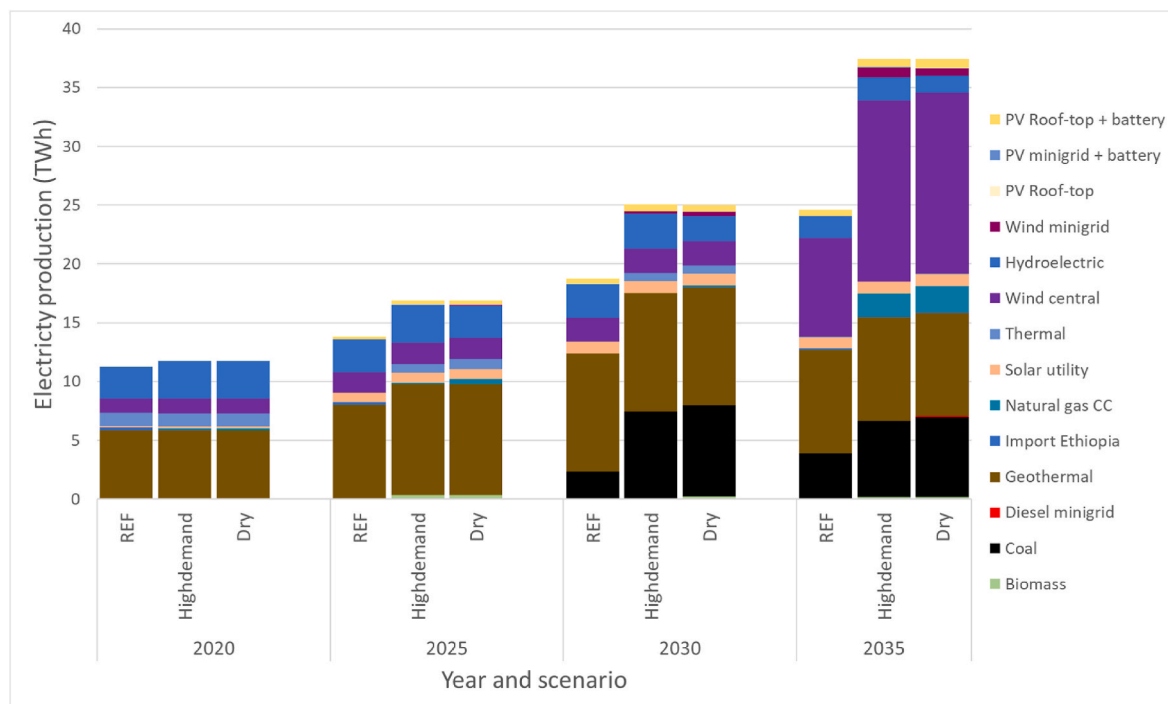


Fig. 7. Electricity production for grid- and off-grid power sources for Kenya for the years 2020 through to 2035 disaggregated by primary resource for all three scenarios (TWh).

Table 3
County results Narok, Kitui and Turkana.

Parameter	Narok county	Kitui county	Turkana county
Total demand 2030	<i>Electrified:</i> 146 GWh <i>Unelectrified:</i> 21 GWh	<i>Electrified:</i> 133 GWh <i>Unelectrified:</i> 20 GWh	<i>Electrified:</i> 17 GWh <i>Unelectrified:</i> 25 GWh
Optimal supply option	<i>Electrified:</i> Central grid 100 % <i>Unelectrified:</i> 100 % PV panel + battery	<i>Electrified:</i> 100 % Central grid <i>Unelectrified:</i> 100 % PV panel + battery	<i>Electrified:</i> 0.4 GWh Wind mini-grid 17.5 GWh central grid <i>Unelectrified:</i> 100 % PV panel + battery
PV panel and battery sizing – low demand	1.9–2.2 kW PV panel with 7–9 h of battery storage for each kW peak demand.	1.6–2 kW PV panel with 5–7 h of battery for each kW peak demand.	1.6–2 kW PV panel with 5–7 h of battery for each kW peak demand.
New capacity required 2020–2030	<i>Electrified:</i> 8.6 MW <i>Unelectrified:</i> 14.1 MW	<i>Electrified:</i> 8 MW <i>Unelectrified:</i> 11.1 MW	<i>Electrified:</i> 0.6 MW <i>Unelectrified:</i> 13.6 MW
Average marginal cost 2030	<i>Electrified:</i> 4 cents/kWh <i>Unelectrified:</i> 9 cents/kWh	<i>Electrified:</i> 4 cents/kWh <i>Unelectrified:</i> 8 cents/kWh	<i>Electrified:</i> 4 cents/kWh <i>Unelectrified:</i> 8 cents/kWh

open-source GitHub workflow can be further enhanced by other users, allowing them to report and track issues. The GIS data used to perform the analysis are listed in text files and all URLs are also listed in Appendix A Table 5. The data is licensed under permissive licences and are in formats that are interoperable such as CSV and shapefiles.

Table 4
Results from the optimisation of PV battery.

	LOW demand profile		HIGH demand profile	
	PV size factor (kW)	Battery hours (kWh)	PV size factor (kW)	Battery hours (kWh)
Median	1.7	5.9	4.5	14.0
Min	1.3	4.7	3.5	10.1
Max	2.5	9.4	7.2	23.9

2.2. Model description - GEOSeMOSYS

In this section, we describe the general modelling steps that enable linking geospatial data with OSeMOSYS, a workflow we call *GEOSeMOSYS*.⁴ The modelling is composed of a computational workflow of pre-processing steps in GIS which identify location-specific data.

We use OSeMOSYS as it enables us to model the central power plants simultaneously with the location-specific data using an LP approach. OSeMOSYS is a multi-year optimisation model that assesses the capacity expansion of technologies and operation of those technologies within a year, using time slices. OSeMOSYS is an open-source ESOM [32] which has been applied to numerous applications including large continental models, and interlinkages in the climate, land and energy sectors and the water nexus [33].

The geospatial information is preserved in the optimisation to be able to explore trade-offs between supply and demand across a country divided into a set of regions simultaneously. In most linear programming environments, there is a limit to how many sets/indices you can define before the problem is not computationally possible to solve. Therefore, the geospatial and temporal data is aggregated to a lower spatial and temporal resolution for optimisation. While this creates simplifications, nonetheless, it provides insights. In this paper, the lower spatial resolution is referred to as *cells* (which are ~40 by 40 km). The higher

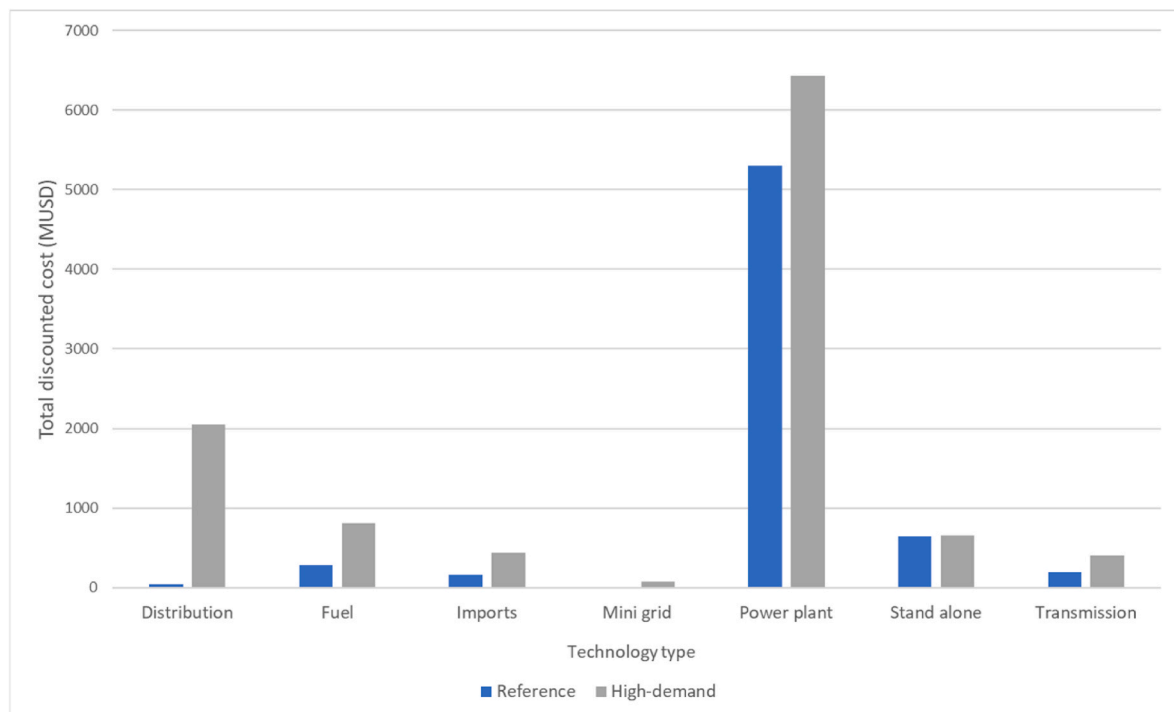


Fig. 8. Total discounted investment and operational and maintenance costs by technology type for reference scenario and high demand scenario 2020–2040.

⁴ https://github.com/KTH-dESA/GEOSeMOSYS_Kenya.

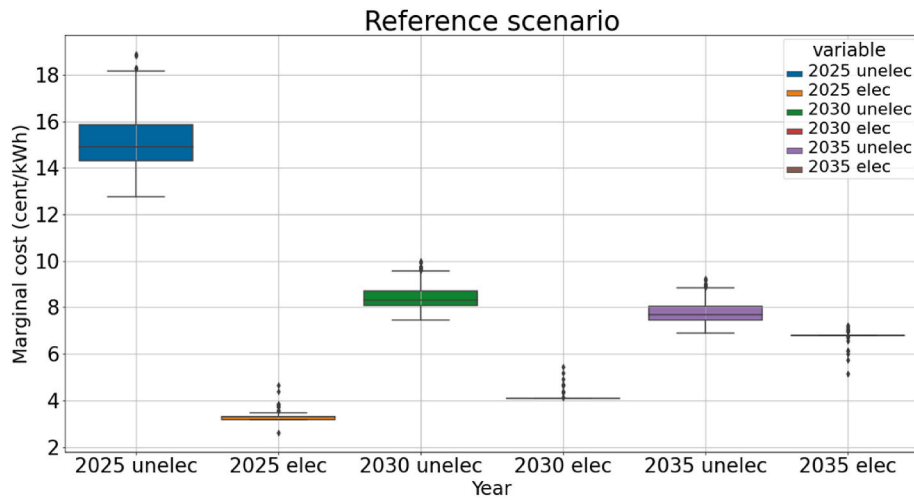


Fig. 9. Box plot of the marginal cost for reference scenario for all 374 unelectrified demand cells (unelec) and 217 electrified cells (elec) for the years 2025, 2030 and 2035.

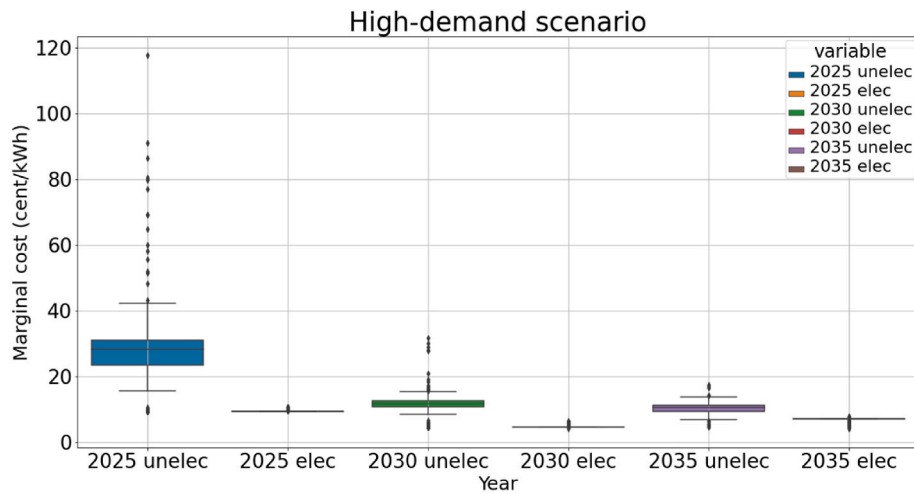


Fig. 10. Box plot of the marginal cost for the high-demand scenario for all 374 unelectrified demand cells (unelec) and 217 electrified cells (elec) for the years 2025, 2030 and 2035.

resolution (1 by 1 km) is referred to as *settlements*. The choice of 1×1 km was chosen as it is considerably higher than the cells, and a convenient unit for running the GIS calculations. The analysis of settlements enriches the information for each of the cells.

The geospatial processing is first handled in a higher resolution for three calculations as depicted in Fig. 1:

- 1) estimating where the electrified population live,
- 2) estimate the distance of low voltage (LV) lines to un-electrified population,
- 3) socio-economic demand indicators.

The following subsections describe the general steps to build the OSeMOSYS data file and model file.

2.2.1. Estimate current electrified cells (1×1 km resolution)

Electricity access is not binary, and depending on the appliances and service level that is required, this leads to very different demands and demand profiles [34]. However, a division between electrified and un-electrified is essential to understand the current electricity infrastructure.

To do this we run an *electrification algorithm* [18] which

approximates which settlements are electrified in the base year. Night-time light is a good proxy for identifying where households have access to electricity [35]. In addition to night-time light, other datasets are considered such as distance to substations, low voltage (LV) lines, medium voltage (MV) lines, and existing mini-grids.

In many cases, data availability is limited in developing countries, however, some global datasets are available. The GIS data that is used for all GIS processing in this paper is listed in Table 1. Some of the data are global, and some are specific to Kenya. However, they can be changed to other regions specified by the user.

2.2.2. Demand distributed to electrified/unelectrified cells

The demand cells are divided into two types, electrified⁵ and un-electrified,⁶ to allow separate demand profiles and projections. This gives a more detailed view of the remote areas which are not connected to the grid in the base year. To approximate the electrified demand for each demand cell the GDP PPP [43] and population are used as a proxy to distribute the demand. A higher GDP is assumed to have a higher

⁵ In base year. This is implicit throughout the paper.

⁶ In base year. This is implicit throughout the paper.

energy activity, thus high GDP PPP will give a higher share of demand [46].

2.2.3. Transmission lines to un-electrified cells

An extension of the transmission line to unelectrified cells is considered in the model. For cells that lack a grid connection in the initial electrification analysis (e.g., are not electrified or are electrified by a mini-grid), the distance to extend the transmission lines is calculated for nearby cells. Cells located further away than 50 km can only be connected by either an existing or newly electrified adjacent cell. This limit was set because the resolution is 40 by 40 km, to only include neighbours to reduce the number of combinations. A decision flow chart of how the adjacent cells is connected can be found in Appendix B in Figure 13.

To enable the adjacent transmission lines to supply the next transmission line they are connected cell-by-cell in the model. This enables the model to expand to adjacent cells from the existing grid without having overlapping lines extended in parallel. In addition, nearby mini-grid cells can supply adjacent cells as they can supply the connecting transmission lines.

2.2.4. Distribution lines to unelectrified cells (Pathfinder)

To identify the required distance of the new distribution lines to connect the unelectrified settlements, an approximation using a greedy algorithm of many-to-many Dijkstra algorithm is applied. The original Dijkstra algorithm finds the shortest path between two points [47] however, a network is not a two-point problem.

The many-to-many Dijkstra algorithm was developed by Roher at Facebook [48] and keeps the shortest path network in a heap. The Pathfinder algorithm takes three matrices as inputs: *origin*, *targets*, and *weights*. The origin is the starting point for the network. The targets are to be included in the network. The weights are the “cost” of travelling that route where low weight is cheaper to follow and will many times be preferred. The Pathfinder algorithm has been applied to identify existing MV lines where data availability is low [49] with good accuracy (75 %) compared to known existing MV lines.

MV- and LV lines follow roads due to logistical reasons, and therefore the weights are lower following the road (weight 0.13). Additionally, the current High Voltage, MV and LV lines are included since that is existing infrastructure (with a weight of 0.00001). The remainder of the weights are set to 1 [49].

The *targets* in the algorithm are the unelectrified settlements from the electrification algorithm. The shortest path network distance is then reduced where there is an overlap with the existing grid to avoid double counting.

To account for additional distribution lines per 1×1 km cell the gridded length from Ref. [50] was adapted for the model and aggregated for an average for each cell (for details on equations see Appendix C Equation 5-10).

2.2.5. Distribution lines in OSeMOSYS to unelectrified in the base year

To respect the spatial dimension in OSeMOSYS, but at the same time allow varying cell sizes, we introduce a new variable *km* in the OSeMOSYS model which represents the newly installed distribution lines length.

Within each cell, the maximum total length of the installed distribution line allowed is the estimated length from the distribution lines length estimation in 2.2.4. The peak demand for the unelectrified cell is calculated per year and accounts for losses. The peak is then divided by the number of km calculated for the cell. The parameter *Peakdemand* represents the unit peak capacity/km and is defined as an exogenous parameter (for details see Appendix D Equation 11 and, Equation 12).

This gives a uniform peak load, which in case the full demand is met by distribution lines, will require the system to install all estimated lengths of distribution lines. The *Peakdemand* increase is proportional to the annual max peak which leads to the distribution line cost can then be

defined as cost per km. The algebraic formulation can be found in Appendix D in Box 1, 2, 3 and Constraints 1 and 2. All parameters and equations added to the OSeMOSYS model file can be found on the GitHub page for [GEOSemOSYS_Kenya](https://github.com/KTH-dESA/GEOSemOSYS_Kenya).

2.2.6. Intermittent electricity generation and storage

For the solar and wind potential the hourly location-specific data is retrieved from Renewables Ninja. The underlying data from MERRA-2 is in resolution 50 km [51] and then interpolated between the points. The advantage of the dataset from Renewables Ninja is that it gives an hourly approximation for each location [44,45]. In the model, the mini-grid system is modelled as a standalone system which is then connected with distribution lines to the demand in the same cell. The wind turbine used in the model is Vestas 42, for the time interval 1st of January to December 31, 2016. This turbine was chosen as it has a rated capacity of 600 kW allowing for a lower capacity range in rural settings. For the Solar PV, the tilt was set to 4° [52] and the azimuth to 180° ⁷ for all solar PV modules. The 2016 wind and solar profiles are assumed to be representative for all years up to 2035 and not sensitive to climate change. This is a simplification and variability over years does occur where some periods are cloudier than others and wind fluctuates, however, as the model is modelled at 12 time slices, these are less pronounced than in a high temporal resolved model as intraday fluctuations are averaged out. On a global level, the capacity factors for solar PV and wind between 2000 and 2017 only fluctuated up and down a few percentage points [53].

For PV panels with batteries, a pre-optimisation is performed to size a large enough PV panel and battery to cover an hourly load profile for a user-defined period. The optimisation input data is a load profile with a peak demand of 1 kW, the PV panel is sized at 1 kW. The objective function, *F* in Equation 1, is to minimise the cost-optimal mix of PV panel and battery hours, subject to constraints where all load is met for the (user-defined) modelling period.

Equation 1. Objective function PV battery optimisation

$$F = PV_{size\ factor} \bullet PV_{cost} + Battery_{max} \bullet Battery_{cost}$$

where $PV_{size\ factor}$ is a factor to size the PV panel (kW), the PV_{cost} is the cost per kW, $Battery_{max}$ is the maximum energy (kWh) stored in the battery and $Battery_{cost}$ is the cost per kWh.

Equation 2. Battery charging and discharging [54]

$$E(t) = E(t-1) + \left[Pch(t)\eta_{ch} - \frac{Pdis(t)}{\eta_{dis}} \right]$$

$$E(t) \geq 0$$

$$E(t) \leq Battery_{max}$$

The battery state, $E(t)$, in Equation 2 equals the previous state of the battery $E(t-1)$ for time t , plus the available power surplus from the PV panel to charge the battery $Pch(t)$, and the power deficit $Pdis(t)$ when the power is supplied by the battery. The efficiencies for charge, η_{ch} , and discharge η_{dis} are set to 0.95 and 0.98 [54]. The full set of equations for the optimisation is detailed in Appendix E and on GitHub⁸ (adapted from Ref. [55]).

To size the different systems, optimisation is applied to each of the 378 cells to ensure enough storage level throughout the sample for each location and load profile. The location-specific optimal PV panel size and battery hours are then multiplied by the capital cost in the model. The demand profile and capacity factor are optimised exogenously and therefore the modified capacity factor for PV panel with battery, when

⁷ Azimuth angle = 180° (i.e., south-facing panels) for farms in the northern hemisphere. Kenya is situated on the equator, in both the north and south hemispheres.

⁸ https://github.com/KTH-dESA/GEOSemOSYS_Kenya/blob/master/src/PV_battery_optimization.py.

accounting for the size of the PV array, is set to 1.

The modelled off-grid technologies are *mini-grid*: wind, diesel generator and PV (with battery) and *stand-alone*: PV (with and without battery). The solar, wind and diesel mini-grid options are designed with distribution extension to the demand sites (Appendix A, Figure 14). All mini-grid installations are also connected to the adjacent cells and can act as a local supplier at the cost of building the connecting MV line and new distribution lines.

2.2.7. Marginal cost in linear programming

From an LP optimisation, the *dual value* for each of the constraints in the model can be obtained. The dual value is the marginal change per one-unit increase and is often called the *shadow price* or *marginal cost*. The shadow price represents the marginal cost of the modelled commodity, in this paper, electricity [56]. This is not the same as the Levelized cost of electricity (LCOE) which typically represents the net present cost of electricity generation for *one technology* over its lifetime. In contrast, the dual value includes all parameters (multiple technologies and fuels) that the cost-optimal additional unit of electricity would cost for that specific demand. This information is very useful to understand the local costs under complex supply chains, where the demand is supplied by several technologies.

2.3. Case application - Kenya

We use Kenya as a case application for two reasons: first Kenya has still a large share of the un-electrified population (~30 %), and second, there are detailed open-access geospatial data from which the model could build.

To model Kenya, we define some initial structural spatiotemporal settings which will be leading the remainder of the modelling effort. The modelled spatial resolution of 40 by 40 km cells leads to 378 cells for the country. To be able to model seasonal changes in supply and demand the model has 4 seasons (January–April, May–July, August–October, November–December) and three day-splits with a peak demand in the evening between 19 and 22, and the break between day and night at 6 a. m. The modelling period is 2016–2040 to calibrate the starting year and to avoid edge effects in the final years.

For Kenya, we have applied the same discount rate as in the *Updated Least Cost Power Development Plan 2020–2040* (LCPDP)⁹ at 12 % [57,58]. Kenya is endowed with large geothermal, solar, wind, as well as hydropower resources. The generation mix in 2020 was supplied 93 % by renewable technologies. The potential power plants: Biomass, Coal, Geothermal, Hydroelectric, Import from Ethiopia, Natural gas CC, Nuclear, Solar utility, Concentrated solar power, Thermal (medium speed diesel and heavy fuel oil generator), and Wind power to supply the grid. In addition, they are harmonised with the planned power plant expansion (Appendix A Table 8) as well as the estimated potential for each technology [57]. Detailed techno-economic data for power plants can be found in Appendix A in Tables 6, 7, 8 and 9.

2.3.1. Estimate current electrified cells (1 × 1 km resolution)

In the case of Kenya, the night-time light (with a value over 0.5) [59] identifies 38 % of the population as electrified in combination with the High-Resolution Settlement Layer (HRSL) [37]. To identify the remainder of the electrified households (70 %) [29] additional parameters such as existing mini-grid and transmission networks are added to the electrification algorithm. As seen in Fig. 2 the estimated electrified population is centred around the south-/southwest of the country.

⁹ The LCPDP is a regularly updated report developed by all sector utilities in the Kenyan power sector. It includes e.g., an overall analysis of demand evolution, and existing and planned expansions of power plants and transmission lines.

2.3.2. Demand distributed to electrified/unelectrified cells

Based on the electrification algorithm, a total of 591 electrified and unelectrified *demand cells* are identified in Kenya. The total demand in the model follows the overall projected demand from the LCPDP 2020–2040 for the reference and vision scenario [57]. The LCPDP include a reserve margin of ~19 % which is not included in the scenarios for the case application.

For the unelectrified, the reference demand follows the estimated rural demand from LCPDP of 78 kWh/connection and year. However, in the *High-demand* scenario for unelectrified, the demand is increased significantly to 1000 kWh/connection and year to juxtapose the technology choice for unelectrified cells in GEOSemOSYS. The reference scenario was chosen to reflect the projected development in Kenya. The other scenario was added as diagnostic scenarios to understand the behaviour of the model under high demand.

In Kenya, electricity consumption was 2019 32 % in residential and 68 % in other sectors [60]. In the LCPDP the residential demand is described to be driven by demographic factors, therefore the residential demand is assumed to be coupled with population density. The industry is in the LCPDP coupled with GDP and is therefore assumed to be driven by the GDP distribution [57,58] as seen in Equation 3 and Equation 4.

Equation 3. Electrified cells demand distribution

$$\text{Electricity demand}_{\text{electrified cells}} = \left(\frac{\text{GDP}_{\text{XY demand cell}}}{\text{GDP}_{\text{sum Kenya}}} \cdot 0.68 + \frac{\text{Population}_{\text{XY demand cell}}}{\text{Population}_{\text{sum elec}}} \cdot 0.32 \right) \cdot \text{demand}_{\text{elec}}$$

where the $\text{GDP}_{\text{XY demand cell}}$ is the total GDP for the specific cell which is divided by the total $\text{GDP}_{\text{sum Kenya}}$ for Kenya. The $\text{Population}_{\text{XY demand cell}}$ is the total sum of the population in that demand cell, divided by the sum of the electrified population in Kenya. The sum of these fractions after multiplying by their respective weight is then multiplied by the $\text{demand}_{\text{elec}}$ which is the total demand for the electrified sector.

Equation 4. Un-electrified cells demand distribution

$$\text{Electricity demand}_{\text{unelectrified cells}} = \frac{\text{Population}_{\text{XY demand cell}}}{\text{Population}_{\text{sum unelec}}} \cdot \text{demand}_{\text{unelec}}$$

The electricity demand_{unelectrified cells} are the demand per cell for the unelectrified. The $\text{Population}_{\text{XY demand cell}}$ is the total sum of the population in that demand cell, which is then divided over the sum of the unelectrified population in Kenya. This fraction is multiplied by the $\text{demand}_{\text{unelec}}$ which is the total demand for the un-electrified sector.

To still be able to compare the results to the Vision scenario from the LCPDP 2020–2040 the unelectrified demand is deducted from the overall projected demand. This means that the electrified demand is lower than projected by the government for this scenario, as can be seen in Fig. 3.

The demand profile for the electrified (217) demand cell is assumed to have the same profile as the half-hourly demand profile for Kenya as a whole in 2015 (Fig. 4) [62]. The initial demand profile for the unelectrified demand cells is assumed to not be the same as the grid-connected cells in the *reference scenario* due to the low demand per connection. As a reference, the total unelectrified demand for 2030 amounts to 0.44 TWh and the electrified 14.75 TWh.

A rural demand profile is not readily available for Kenya. However, a study from Tanzania on rural microgrid smart meters is assumed which analyse different customer segments in microgrids (2018). To be able to depict an average rural customer in all types of premises (i.e., home, business, and public premises) they are aggregated into one low-demand profile in the model. As seen in Fig. 4 the low-demand profile has a clear peak in the evening, whereas the high demand has a lower peak with a flatter demand over the day.

2.3.3. Transmission lines to un-electrified cells

For Kenya, 656 transmission lines between unelectrified cells are

assessed in the optimisation. The relation between cost and capacity is assumed to be linear in the model. The transmission cost per kilometre of installed transmission line per MW of power capacity delivered is 3300 USD and the cost for the substation is set to 24000 USD/MW [63]. The cost per kilometre of installed transmission line per MW transmission line is multiplied by the length of each line which gives a cost per capacity unit. In addition to this, there is a central cost per installed capacity to account for additional costs related to the expansion of the transmission lines of 365 USD/kW [64]. The transmission losses are 4.5 % in the base year and decrease to 3.5 % in 2025 due to expected efficiency measures. Distribution losses are modelled at 17 % [57,58] for the whole modelling period.

2.3.4. Distribution lines to un-electrified cells (Pathfinder)

The total distance from the Pathfinder algorithm, to estimate the shortest network for distribution lines, amounts to 132,938 km (disregarding traversing distances). Accounting for the average needed length within each settlement the total length of the distribution lines to connect all unelectrified cells amount to 328,392 km. The current length of distribution lines in Kenya is 76,319 km (of lower voltage type 11 kV and 33 kV in 2020) [57,58]. For the additional length of the LV lines per cell, the average length ranges between 0.44 km–11 km¹⁰, which is then multiplied by the estimated network length from the Pathfinder algorithm.

2.3.5. Intermittent electricity generation and storage

To capture learning curves the projected costs are taken from NREL Annual Technology Baseline (ATB) [65]. However, the overall residential battery cost considered in the ATB considers the labour cost in the U.S. As the labour cost is not as high in Kenya the labour cost is reduced for residential batteries as it represents a very large share of the total cost. According to the Global Wage Report, 2021–2021 [66] the ratio between the highest minimum wage in Kenya compared to the U.S. was approx. 0.11. Therefore, the *labour cost* in the ATB forecast, for residential batteries, is reduced by that factor. Similarly, the rent and building cost was assumed to have the same proportional reduction as labour. In total, this leads to a range of battery costs in 2030 of US\$ 2003 in the advanced scenario and US\$ 2845 in the moderate scenario for a 4 kWh battery. Comparing these prices [67,68] both conducted off-grid studies in African countries and they modelled the capital cost of batteries at \$US 300/kWh. A study in India [10] modelled the cost at \$USD 340/kWh. All three studies have lower costs than the reduced capital cost from NREL. The NREL cost does, however, include e.g., the installation cost, overhead, and permits which might not be included in the other studies.

The PV panel and battery is optimised between 1 and 31st of January to ensure a high service level over a longer period. This is a higher service level than the MTF propose where the reliability at Tier 4 is a maximum of 14 disruptions per week and Tier 1–3 does not have any stated reliability [34]. The costs for the optimisation was set to the 2021 value of 2508 USD/kW for PV and 522 USD/kWh [65].

2.3.6. Scenarios

Two types of scenarios are explored in this paper. First, the reference scenario is harmonised with the overall demand projections from the Kenyan government and can be considered a realistic scenario. The second type of scenarios are diagnostic and are used to assess the validity of the model. The first “high-demand scenario” assesses final electricity demand, a key driver for cost and technology choice [69,70]. The “dry-hydroelectric scenario” explores the dynamics of the model when the hydroelectric power availability is affected by climate change [57, 58]. Table 2 summarises the scenarios.

¹⁰ With a peak demand of 200 W/connections, max capacity of the LV line to 10 kW, and an inhibited area of 0.3 across all cells and scenarios [50].

2.4. Computational requirements

The detailed demand curve combined with geographical information and many decentralised technologies leads to a large LP model which consists of 3812 technologies and 977 fuels. The optimisation took ~15–21 h in total to run depending on the scenario on the High-performance computer (HPC) Dardel, KTH.

3. Results

In this section, the results from the modelled scenarios for the case application for Kenya are described concerning the electricity generation, costs, emissions and capacities.

3.1. The role of demand

For the unelectrified population in the reference scenario, all of the newly electrified demand is met by residential PV panels with batteries. An expansion of the connecting transmission lines in the northern part of Kenya is seen in the results, however, only to electrified cells where the distribution network to the households already exists. In Fig. 5 a and b the expansion of the transmission lines is displayed for 2030, and 2035 in combination with the installed new distribution lines. The energy flow in the connecting transmission lines ranges from 1 to 605 MWh per year.

For the high-demand scenario, the supply options are instead the opposite with 91 % of the unelectrified demand supplied by either mini-grid (~6 %) and/or grid. The higher demand, together with a higher demand share during the night compared to the reference scenario, leads to more diversified supply options. As seen in Fig. 6 a and b the distribution lines are installed already in 2030 for most of the cells. It is observed both where existing transmission lines are already installed in the base year and new extensions with transmission lines to adjacent cells. The transmitted electricity ranges between 1 and 166 GWh per year.

Looking at the installed capacity of powerplants, mini-grid and stand-alone technologies for the different scenarios, the new generation capacity for the reference case amounts to 7.6 GW which is slightly higher than in LCPDP 2020–2040 (which amounted to 5.7 GW) [57]. For the high-demand and dry-hydro scenario, the new capacity amounted to 10.3 GW respectively 10.5 GW compared to 14.5 GW in the LCPDP Vision scenario [57].

Comparing the electricity production between the different scenarios, as seen in Fig. 7, the predominant technology choice is geothermal for all scenarios, and towards the end of the modelling period, wind power also supplies large shares of electricity. The 2020 energy mix is the current electricity generation mix. The planned coal power plant is not fully utilised in the reference scenario, starting at a 4 % utilisation in 2027 and increasing to ~88 % towards the end of the modelling period. For the dry-hydro scenario, a decrease of 14 % in 2025 and 28 % in 2030 in hydroelectric power is observed compared to the high-demand scenario. The decreased hydroelectric power production is replaced by natural gas, coal, wind, thermal power plants, and distributed technologies.

3.2. County-level results for Narok, Kitui and Turkana

To show the results on a county level, we present three sample counties namely: Narok, Kitui and Turkana. The county-level analysis is disaggregated by the population density and only detailed for the reference scenario as the other two scenarios are more diagnostic. Turkana county has many unelectrified areas and covers a large area. Kitui and Narok are also large counties situated in different parts of the country. In Table 3 the different key inputs and outputs from the modelling have been summarised. For Narok and Kitui counties all cells have populations that are electrified in the base year which means there

is transmission connectivity to the county throughout.

In the County Integrated Plans (CIP) for 2018–2022, the electricity access for Kitui was 18.2 % (2015/16 census) and Narok 20 % (2018) [71,72]. The modelling results from GEOSemOSYS for 2020 estimated 40 % electricity access for Kitui and 30 % for Narok. Kenya has had rapidly growing electricity access from 36 % in 2011 to 71 % in 2020 [73] and therefore, a high increase on the county level is expected in just a few years.

3.3. Investment costs and local marginal costs

The total discounted costs of investments and operational and maintenance costs in the reference scenario amount to 6633 million USD (period 2020–2040). The same scenario is estimated to amount to 10,055 million USD for the same period in LCPDP [57]. The cost breakdown seen in Fig. 8 is for the largest share represented by the cost for power plants and then followed by stand-alone technologies. The high-demand scenario's total discounted cost amounts to 10,870 MUSD and the dry-hydro scenario's total discounted cost amounts to 10,986 MUSD for the same period. The cost breakdown for the high-demand scenario is also predominantly for power plants followed by extending the distribution lines.

The share of transmission in the total capital cost for both power plant and transmission in the high-demand scenario was 10 % and in line with what [63] estimates as between 4 and 15 % of the total cost.

The PV panel and battery pre-optimisation for the 378 cells in Kenya will, for the low demand profile led to a median of 1.7 kW and 5.9-h battery, and for the high profile scenario a median of 4.5 kW and 14-h battery as seen in Table 4.

Looking at the weighted annual average undiscounted marginal cost in Fig. 9, for the reference case, a few patterns appear. The first pattern is that in the initial year of 2025, the marginal cost is stable at a median of 15 cents/kWh. In 2030 the range decreases to a median of 8.5 cents/kWh. The demand profile for unelectrified has a very high evening peak in the reference case and all the demand is supplied by PV with batteries. The marginal cost is zero for all time slices except for the evening peak for January–April. In practice, that indicates that in all other months and times of the year, there is an oversupply compared to the modelled demand.

For the electrified, the cost is more stable with a median below 4 cents/kWh in 2025 with slightly increasing costs over time when moving up the supply curve.

Comparing the marginal cost for the high demand, seen in Fig. 10, a large interquartile range with a median of 28 cents/kWh is seen in 2025. This illustrates the variation in supplying technologies. Here the demand profile is flatter throughout the modelling period, however, the technology supplies are more versatile, and therefore higher marginal costs are observed in overall time slices. The outliers in both scenarios show the diversity of supply options in these scenarios.

3.4. Carbon-dioxide emissions

Kenya has in their Nationally Determined Contribution (NDC) to the Paris agreement promised a reduction of 30 % compared to their business-as-usual scenario which is equal to a total of 100 MtCO₂e for all sectors. Considering that the electricity sector has close to zero emissions in the base year, and the other sectors' emissions are expected to increase, the electricity sector needs to be kept at the same or lower levels of emissions to meet the NDCs [74]. The carbon-dioxide emissions from start at around 0.3 MtCO₂ and steadily increase to a maximum of 2.3 MtCO₂ in the reference scenario and over 3 MtCO₂ for high-demand and dry-hydro scenarios. The planned coal plant is installed in 2027 which has a larger share of production in the high demand and dry-hydro scenario which emits more CO₂. In the LCPDP 2017–2037 (as the emissions are not reported in the 2020–2040 report) the reference case amounted to 0.3 MtCO₂e in 2030 and the Vision scenario to 4 MtCO₂e in

2030.

4. Discussion

In this paper, we have explored how to increase both the spatial and temporal resolution in an ESOM for energy access. We used Kenya as a case application and showed that in the reference scenario disaggregated demand leads to mostly off-grid connections for an unelectrified population. For our diagnostic scenario, with the high demand, almost all demand was connected to a mini-grid/grid. Other studies have shown similar results where low demand in spatially disaggregated models leads to more off-grid solutions while higher demand leads to more mini-grid or grid connections [17,18,21].

The marginal cost for Kenya gives insights into of the cost of electricity for each of the 591 cells. The range was quite large in the scenarios ranging between 1 and 120 cents per kWh. This raises questions about affordability when the marginal cost is high. The trade-off between supply and demand was clear, especially when the supply technology choice was PV panels with batteries in the reference scenario. The service level required by the PV panels with battery was high and led to higher capacity with zero marginal cost in all time slices other than the evening peak. This was also observed in Ref. [10] where hybrid systems with PV panels, batteries and diesel generators were explored to not oversize the PV panel and battery. This indicates the importance of appliance mixes associated with electrification programs. The less 'peaky' the appliance-specific demand, or the more that the appliance can effectively 'shift' demand, the lower the requirement for centralised grid rollout - and the institutional burden that can bring.

We presented aggregated results for different counties. Furthermore, insight into the planned expansion plans and the power surplus that was observed for the coal power plant in the reference scenario was highlighted. The GEOSemOSYS model set-up can therefore provide insight into these trade-offs. However, the modelling approach has both insights and trade-offs, and aspects of these need to be discussed to highlight these for energy planners.

The methods of GEOSemOSYS apply not only to Kenya but to studies that want to understand technology choices for energy access and the grid-supplied power plants in one energy systems model.

Important key spatial aspects are captured in the spatially disaggregated model which a "one node" (e.g., on a national level) model cannot capture. First, the trade-off between supply and demand in a location-specific context provides insight into the required service level. The marginal cost can give further insight into the cost of the service level in a complex energy system. Second, the model handles both the central power plants, expansion of the transmission and distribution lines and distributed technologies in one multi-year model.

Third, the model is built with open-access data and open-source code. The open-source code can be further enhanced by other users, and issues reported to further improve the workflow. The chosen spatial resolution leads to a large number of technologies and fuels. However, the model can be solved on a desktop computer with 32 Gigabytes of Random Access Memory if the fast version of OSemOSYS is used. Post-processing is then needed to calculate some of the results. Open-source solvers are available such as CBC (COIN-OR Branch-and-Cut solver) which can be used instead of IBM ILOG CPLEX Optimizer, which is only free on an academic license. The model generation code is also largely programmed in Python and the spatial resolution is relatively flexible, so it is possible to reduce the number of cells and run a lower resolution on a desktop computer.

Fourth, the spatial dimension is better captured with the Pathfinder algorithm to estimate the required distribution lines to unelectrified, which is a slightly different application than the previous version which looks to estimate the existing MV network [49]. This together with the added constraints allows OSemOSYS to incrementally install distribution lines supplying the final demand.

However, several key parameters in the model have high

uncertainty. The electrification algorithm from which estimations on where the electrified and un-electrified population live is a key assumption for the analysis. This estimation is difficult to validate and is central to the remainder of the calculations. A study by Ref. [35] found that, compared to the available census from the study countries, night-time light is broadly a good parameter to approximate the electrified population. In our study, we included several more parameters from the electricity infrastructure (e.g. transmission lines, substations, transformers) where the distance to the nearest road, transformer or low-voltage line were the main parameters in combination with night-time lights, a detailed breakdown of the decision criteria for electrifying a cell can be seen in Appendix B in Figure 12.

All the spatial data is located in the centre of the cell meaning that the solar and wind capacity factor and grid extension is at the centre of the cell. As the cells are coarse (40 by 40 km) better and worse conditions would certainly exist within the cell. In addition, the reserve margin is not included as the computational time increases dramatically when modelling with high spatial granularity.

There are also limitations to the model relating to the LP approach. The expansion of the transmission lines between the cells is modelled at 3300 USD/MW-km, with no lower limit as LP requires continuous variables. This leads to transmission lines being installed at very low capacities at a low cost per km. This can lead to less installed mini-grid and off-grid options. In addition, the cost is applied in both directions. However, the cost which was added to the existing central transmission system acts as an extra cost per added capacity anywhere in the network, and the total cost for the transmission lines for the high-demand scenario are within the expected range. The spatially explicit connection cost of new power plants is not accounted for in the analysis, as the location of these are not known. This can lead to higher costs, depending on the location and transmission capacity at the proposed location. However, as there is a general cost of adding more capacity to the existing transmission lines, costs are accounted for on a general level.

The adapted OSeMOSYS model distribution constraints work for the final technology supplying the demand as the peak demand is linearly proportional to the length of installed distribution lines in those cases, as it is the technology that supplies the demand. The transmission lines are however more complex, supplying adjacent cells with multiple demands downstream. Therefore, the same equations cannot be transferred upstream with the current formulation, as it is situated in the middle of the energy system with multiple optimal options. Due to the linearity in LP, the optimal results can install a very small capacity per km, which might be unlikely when planning the electricity system. One way to mitigate this is to add a lower limit and optimise using a MILP. However, the computational time increases dramatically, which was undesirable in this study. Furthermore, the actual power flows are not modelled, so the optimisation is a techno-economic assessment. Finally, the double distribution lines, one supplied from the grid and the other supplied from a mini-grid, in cells that have transmission lines in the base year lead to higher costs as the distribution lines cannot be used concurrently.

5. Conclusions

Planning energy access, when the coverage and extent of the existing transmission grid is poor is a spatial problem. Alternatives to grid expansion include installing distributed technologies. In this paper, we presented an ESOM with a high spatial resolution with open-source model generation code for OSeMOSYS, a workflow we name GEO-SeMOSYS.¹¹ Many ESOMs capture medium to long-term temporal investment dynamics but represent spatial realities poorly. Often, spatially explicit models do not capture this medium-to long-term system-wide dynamics, while in our modelling we capture elements of both simultaneously.

Focusing on the first research question: What is the cost-optimal techno-economic solution for Kenya when representing both spatially explicit decentralised supply and demand and centralised power supply in one ESOM?

In the reference scenario, electricity access is supplied by solar PV panels with batteries, which leads to very low expansion of the transmission grid, at low levels of demand. The costs vary over time and by electrified or unelectrified demand. The cost increased in combination with a sharp evening peak or a high night-time demand when supplied by PV and battery. The marginal cost identified the value of shifting demand. Where it is possible to reduce the sharp evening peak or higher night-time demand profile of new connections the economics of stand-alone systems would be improved (as peak marginal costs would reduce). That might include integrating very high-efficiency appliances; encouraging changes in appliance time of use; or improving storage. This could be electrical, thermal, or chemical depending on the service that is being demanded at its peak.

The diagnostic scenarios show that when supplying a high demand, there is a clear shift to grid and mini-grid supplying the previously unelectrified population. The expansion of the transmission lines can be overrepresented due to the LP approach and overestimate the grid connection. The overall cost structure for the reference scenario has a larger share of standalone compared to the high-demand scenario which instead has a larger share of distributed generation. The majority of the costs in both scenarios are from power plants.

We presented model results on the demand and supply options for three sample counties: Kitui, Narok and Turkana. However, all counties can be synthesised from our results. At the same time, the modelling approach allowed to understand the cost-optimal solution including the spatial dimensions and the latent demand of unelectrified population. In the reference scenario during the initial years some of the planned power plants was not fully utilised. Because both spatially distributed and centralised power plants are modelled at a high spatial resolution simultaneously potential trade-offs between them can be observed.

The model is open-source and built on open-access data and government documents. Since it is open-source, energy planners can download the workflow and modify input data and datasets to further explore other scenarios and developments.

Focusing on the second research question: To what degree can an ESOM be extended to represent spatial information and what are the insights and trade-offs from this approach?

To better represent the spatiotemporal aspects in the ESOM, methodological developments were developed for this paper for both estimating the length of new distribution lines as well as optimising the length to supply the demand in OSeMOSYS. Furthermore, a pre-optimisation of the PV panel and batteries was developed for each location.

We use LP as it allows us to find a global optimal solution and can solve large models. The marginal cost allows us to understand the additional cost of one more unit of electricity under complex supply chains, indicating the cost per kWh for each technology in the broader system context. There are several disadvantages when increasing the spatiotemporal resolution in an ESOM. These are linked to the limitations of LP. The cells are relatively coarse (40 by 40 km) but at the same time, the computational time is relatively long, approx. 21 h for the high-demand and dry-hydro scenario. LP comes with limitations in terms of representing how the energy system would practically evolve. Due to the need for continuous decision variables of the model, small incremental installations of e.g., power plants, transmission lines, and distribution lines are allowed. To overcome this issue MILP can be used, however, the computational time would dramatically increase, and, in our case, we would need to reduce either the spatial or temporal resolution to find an optimal solution. Furthermore, the power flows in the network are not assessed, and running a separate power system model would provide further insights to the networks system design.

¹¹ https://github.com/KTH-dESA/GEOSeMOSYS_Kenya.

5.1. Future work

The spatial resolution of 40 by 40 km was chosen so that the computational time did not exceed ~20 h. As [75] highlight, the results in OnSSET change with different population aggregations. This indicates that it would be of value to also investigate different spatial resolutions further in combination with the temporal resolution for OSeMOSYS to understand the sensitivity of the results. To apply this method to other countries for electricity access could be valuable to understand the dynamics between electricity access, centralised and decentralised supply options. Furthermore, to assess the power flows in a power system model would provide more insight to the network system design.

The current GitHub repository code is specifically developed for Kenya; however, the code can be of interest to other energy modellers and can be modified to be more general. This would not require a huge effort as much of the code is already based on the GIS data provided by the user.

CRedit authorship contribution statement

Nandi Moksnes: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization, Project administration. **Mark Howells:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision. **William Usher:** Methodology, Validation, Writing – review & editing, Supervision.

Appendix A

Table 5
GIS Datasets

Name	Resolution	Reference
Nighttime light (VIIRS) (Nighttimelight data average masked, 2020)	15 arc second	[36]
GDP PPP (GDP_PPP_2015_WGS_84)	300 arc second	[43]
Population (HRSL_Kenya)	1 arc second	[37]
Administrative boundaries (ADM: gadm36_KEN.shp)	–	[76]
Road network (Roads)	–	[39]
Low voltage distribution network (11kV transmission Kenya)	–	[38]
Low voltage distribution network (33kV transmission Kenya.zip)	–	[38]
Medium voltage distribution network (66kV transmission Kenya)	–	[38]
High voltage transmission network (132kV transmission Kenya)	–	[38]
High voltage transmission network (220kV transmission Kenya)	–	[38]
Existing mini-grid locations (mini-grid)	–	[40]
Under development mini-grid locations (mini-grid)	–	[40]
Transformers (transformersKenya)	–	[42]
Transmission stations (transmissionstationsKenya)	–	[42]
Substations (primary-substationsKenya)	–	[41]
Zenodo datasets	–	https://github.com/KTH-dESA/GEOSeMOSYS_Kenya/blob/master/src/input_data/zenodo.txt

Table 6
Battery techno-economic data [65] adjusted to Kenyan wages [66].

Type	Capital cost	2019	2025	2030	2035	2040
Residential	1 kW/4 kWh <i>Battery Storage - Advanced</i>	4589	2806	2043	1843	1643
Residential	1 kW/4 kWh <i>Battery Storage - Moderate</i>	4589	3270	2845	2668	2490

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data is available open access.

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Table 7
Techno-economic data

Technology	Capital cost (USD/kW)	Variable cost (USD/GJ)	Fixed cost (USD/kW)	Operational life (years)	Efficiency/Capacity factor	Availability factor	Annual max capacity
Residential PV NREL (National Renewable Energy Laboratory), 2021)	2020: 2710 2030:1004	-	18	30 [77]	Location-specific [45]	-	
Wind mini-grid [65]	2020: 1375 2030:910	-	39	20 [77]	Location-specific [44]	-	
Transmission and distribution	HV/MV: 3.3 USD/kW-km (location specific) [63] LV: 10000 USD/km (location specific) [17] Increasing capacity for existing transmission lines is set to 365 USD/kW [64] and assumed 80 USD/kW additional for distribution capacity.	-	2.5 % of capital cost (location specific) [74]	60 (outside of the modelling period)	HV lines: 95.5 % Distribution: 83 % [57,58]	-	
PV Mini-grid + battery [65]	2020: 3613 2030: 2024	-	2.5 % of capital cost	30 [77]	Location-specific [45] + optimised battery	-	
PV residential + battery [65]	2020: 7040 2030: 3850	-	2.5 % of capital cost	30 [77]	Location-specific [45] + optimised battery	-	
Diesel generators [78]	650	4.2	15	15	25 % (specific fuel consumption) [79]	-	
Import Ethiopia (200 MW, 612 km Kenya) [74]	1300 [74]	19.44 USD/GJ (7 cents per kWh)	-	60	-	85 %	
Wind central [65]	2020: 1375 2030:910	-	40	25	38 % Seasonal variation	-	4.6 GW [57,58]
Heavy fuel oil [57]	1650	1.2	7.1	20	45 %	92 %	
Hydro run of river	2902	1.2	2	50	38 %	98 %	
Hydro dam medium [74]	4089	0.16	18.6	50		90 %	
Hydro large [57]	3798	0.14	15.3	50	Seasonal variation, on average 58 %. Based on the monthly production of Turkwel and Sondu HPP	90 %	
Hydropower: Masinga Kamburu Gitaru Kinadruma Kiambere High grand Falls [57]	High grand Falls: 3819 Start year: 2031 The remaining hydropower is not installing a new capacity.	0.14	15.3	50	All HPP are seasonally based on their monthly production Masinga: 43 % Kamburu: 48 % Gitaru: 40 % Kinadruma: 29 % Kiambere: 60 % Mutonga: 49 % [80] High grand Falls (same as Mutonga): 49 %	90 %	
Geothermal [57]	3500	Binary: 2.9	Binary: 22.1 Flash steam: 151.9	30	Binary: 80 % Flash steam: 90 %	95 %	
Solar utility [65]	2020: 136 2030: 759		2020: 22 2030: 16	25	22 % (peak capacity factor in daytime, low evening and 0 night-times)		
Biomass [74]	3045	2.5	152.3	30	25 %	90 %	
Medium Speed Diesel power plant [57]	1650	1.7	7.1	25	45 %	92 %	
Natural gas, combined cycle [57]	1349	1	18	20	49.5 %	90 %	
Coal power plant [57]	2553 Start year: 2024 Lamu coal power plant 981 MW installed in 2027 [57]	1.4	68	30	39 %	90 %	
Nuclear [57]	6200 Start year: 2031		66.3	50	37 %	90 %	
Concentrated solar with storage [65]	2020: 6507 2030: 4408	3.5	66.8	25	56 % (peak capacity factor Day and evening, low night-time)		

Table 8
Planned power plants (MW) [57].

Power plant type (MW)	2020	2021	2022	2023	2024	2025	2026	2027
PV utility			161.5	90	72.5		80	
Wind		100		50	50	50	90	
Coal								981
Geothermal	58	83		159.4	173.8	70	140	
Hydro			19.9	2.1		6	20	
Import Ethiopia	200							
Biomass			1	28		10		

Table 9
Emission factors

Technology	Emission factor ktCO ₂ /GJ
Heavy fuel oil	7.01×10^{-5}
Diesel medium speed	6.943×10^{-5}
Natural gas, combined cycle	5.029×10^{-5}
Coal power plant	9.05×10^{-5}
Diesel generator (distributed technology)	8.838×10^{-5} [81]

The fuel price assumptions in the model are harmonised with LCDP 2020–2040/2021–2030 [57,58] until 2021 and thereafter the WEO 2020 projections are applied for stated policies and sustainable development scenarios. According to the LCDP heavy fuel oil is estimated at 75 % of the crude oil price. Additional transport costs are accounted for (and included in the figure below) with 7 USD/ton of coal added as well as 13 % of the import price of diesel and heavy fuel oil (Figure 11).

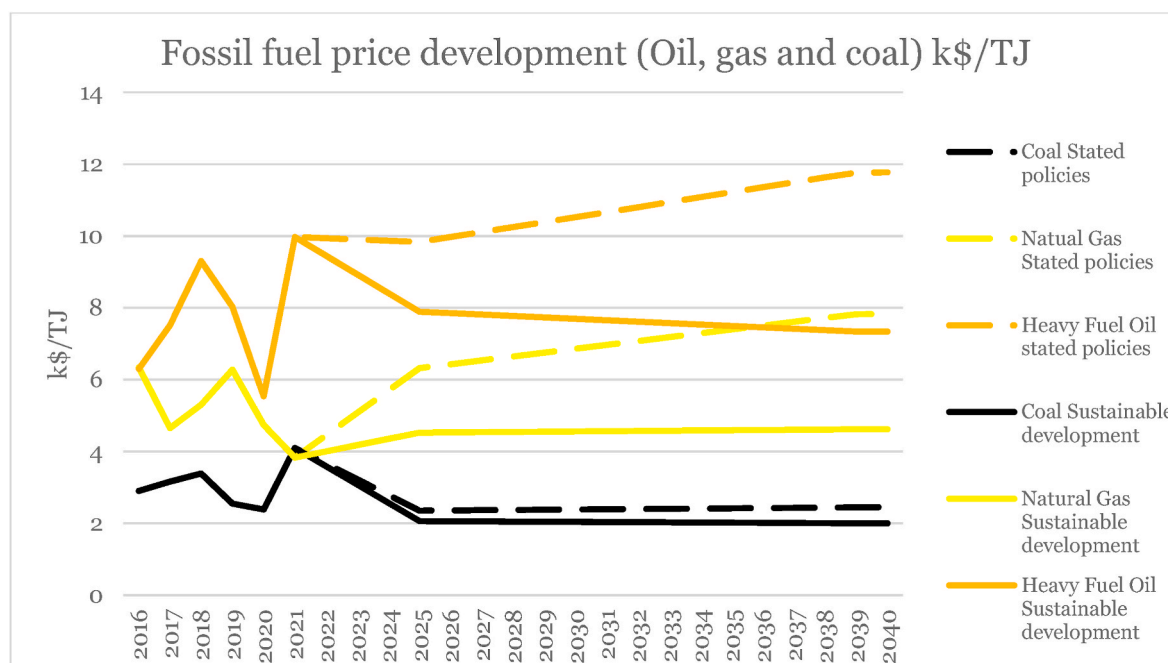


Fig. 11. Fuel price development [57,82]

Appendix B

Table 10
Quality of GIS data used in the study

GIS-data	Compared to literature
Transmission lines:	2019/20:
11 kV: 35,699 km	11 kV: 40,616 km
33 kV: 28,319 km	33 kV: 35,703 km
66 kV: 850 km	66 kV: 1187 km
132 kV: 2558 km	132 kV: 3372 km
220 kV: 1526 km	220 kV: 1686 km
	415/240 V: 158,527 km
[38]	[57,58]
Population (2020):	Population (2020):
53,646,681 people	53,771,300 people
[37]	[83]

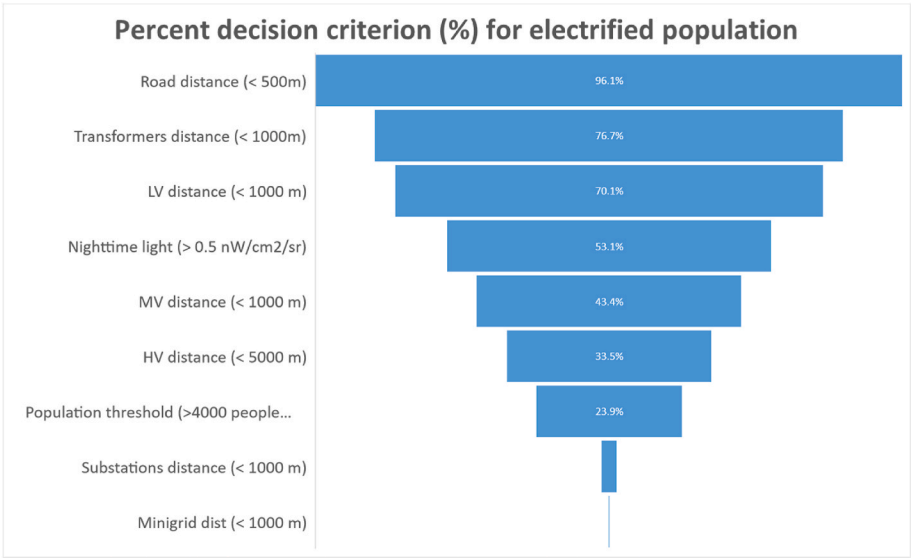


Fig. 12. The electrification algorithm results per decision criterion and settlement

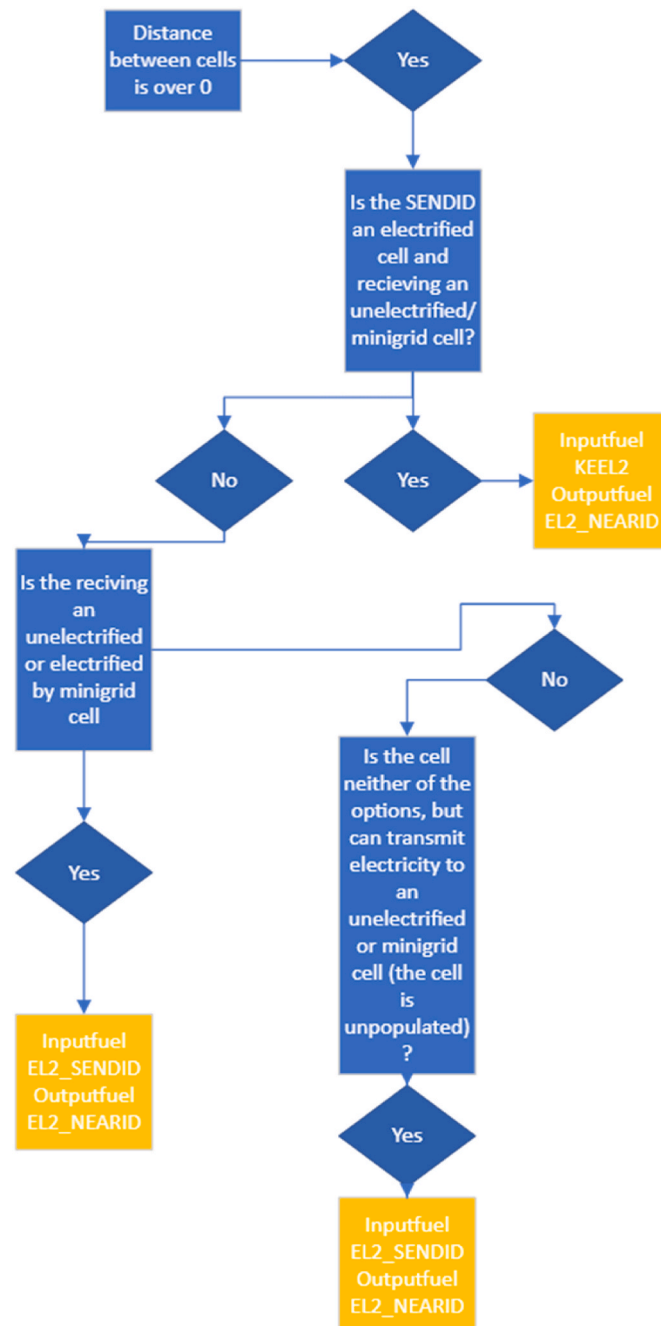


Fig. 13. Decision flowchart of the transmission lines

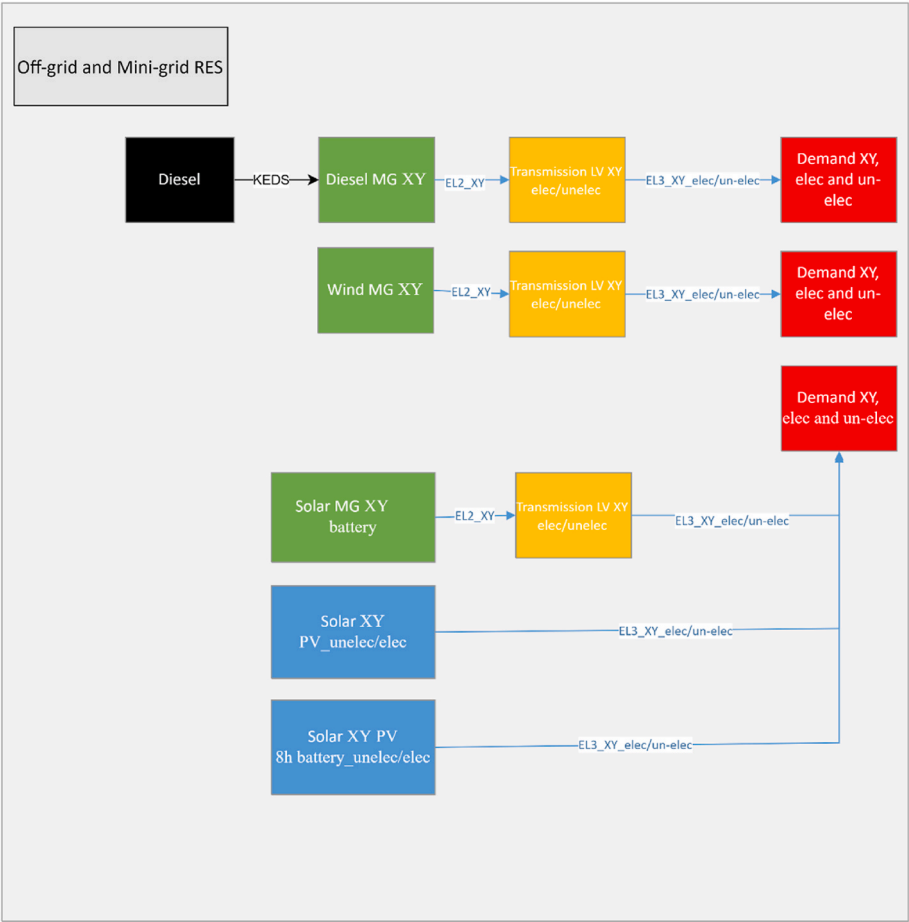


Fig. 14. Off-grid reference energy system

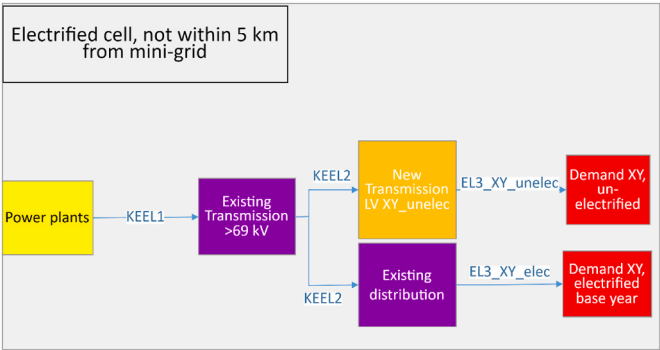


Fig. 15. Central power plant reference energy system

Appendix C

Legend for the gridded Low voltage calculations.

Households = is the number of households per square km.

Peak_{household} = expected peak demand (200 W)

LV_{network max capacity} = the max load in the system (modelled at 10 kW)

Disaggregation factor = Area that is considered inhabited (modelled at 0.3 (Modi et al., 2005)

MaxArea_{LV} = The maximum covered area of LV lines, modelled at 450 km² (the LV maximum length is 30 km).

NrLV_{lines capacity} = minimum LV line capacity

minLV_{line length} = minimal number of LV networks per MV line.

LV_{network per MVline} = the number of LV networks per MV line is the maximum of the required LV networks on the basis of capacity or line length but maximised at the number of rural households in the grid cell.

Household_{LV} = the number of households per LV network.

LV_{network length LV} = the total LV line length per LV network.

LV_{network length} = the length for LV network per cell.

Equation 5 Low Voltage length calculation adapted from [50].

$$NrLV_{lines\ capacity} = \frac{Households \bullet Peak_{household}}{LV_{network\ max\ capacity}}$$

Equation 6. LV line minimum length

$$minLV_{line\ length} = \frac{Area \bullet Disaggregation\ factor}{MaxArea_{LV\ network}}$$

Equation 7. LV networks per MV line

$$LV_{network\ per\ MVline} = \min\{Households, \max\{min\ LV_{line\ length}, NrLV_{lines\ capacity}\}\}$$

Equation 8. Number of households per LV line

$$Household_{LV} = \frac{Households}{LV_{network\ per\ MVline}}$$

Equation 9. Length of LV network

$$LV_{network\ length\ LV} = \sqrt{\frac{Area \bullet Disaggregation\ factor}{Households}} \bullet \frac{\sqrt{2}}{2} \bullet 1.333 \bullet Household_{LV}$$

Equation 10. Length of all LV networks (km)

$$LV_{network\ length} = LV_{network\ length} \bullet LV_{network\ per\ MVline} \bullet No\ MV\ lines\ (assumed\ 1)$$

Appendix D

The algebraic formulation in OSeMOSYS.

This section explains the algebraic formulation that is implemented in the model for GEOSeMOSYS Kenya.

Peak demand calculation

Equation 11. Peak capacity per cell

Box 1

Indices that are used for the equations

 $r...$ Region $t...$ Technology, e.g., PV panel or Coal power plant $y, yy...$ Year [startyear, startyear + 1 +...+ final year]

Peak demand calculation

$$Peak\ max_{xy\ year}(capacity\ unit) = \max \left(\frac{SpecifiedAnnualDemand_{xy,year} \bullet SpecifiedDemandProfile_{ts,year}}{CapacitytoActivity \bullet YearSplit_{ts,year}} \right)$$

SpecifiedAnnualDemand is the annual demand per demand cell, SpecifiedDemandProfile is the demand profile per cell and CapacitytoActivity represents the ratio between energy and capacity. The YearSplit is the time slice share of a year.

Equation 12. Peak demand per cell-km

$$Peakdemand_{xy,year} \left(\frac{capacity\ unit}{km} \right) = \frac{Peak\ max_{xy,year}}{Distributionlosses_{xy} \bullet Estimatedkm_{xy}}$$

Distributionlosses are the losses at the specific location and Estimatedkm is the length of distribution lines per cell (km) (which is an output from the distribution lines calculation).

Box 2

Parameters entered by the analyst

Peakdemand (r,t,y) – The peak demand per cell/km as defined in Equation 7

SpecifiedAnnualDemand is the annual demand per demand cell, SpecifiedDemandProfile is the demand profile per cell and CapacitytoActivity represents the ratio between energy and capacity. The YearSplit is the time slice share of a year.

Equation 12CapitalCostkmkW (r,t,y) – The cost per km of distribution line.

Box 3

Variables added to the model

 $km_{r,t,y}$ – A variable that represents the length of distribution lines that is needed to meet the exogenous peak demand.CapitalCostnetwork $_{r,t,y}$ – The annual cost of distribution line.

Equation 12. CapitalCostkmkW(r,t,y) – The cost per km of distribution line.

Constraint 1. OSeMOSYS distribution model.

(NewCapacityDistribution)

$$\forall_{r,t,y} : Peakdemand_{r,t,y} \bullet \sum_{yy:y-yy \ \& \ y-yy \geq 0} km_{r,t,yy} \geq TotalCapacityAnnual_{r,t,y}$$

Constraint 2. OSeMOSYS distribution model.

(NetworkCostPerCapacitykm)

$$\forall_{r,t,y} : km_{r,t,y} \bullet CapitalCostkmkW_{r,t,y} = CapitalCostnetwork_{r,t,y}$$

Appendix E

PV battery optimisation

This section covers the additional equations that are covered in the PV panel and battery system optimisation.

Equation 13. Available charging to battery

$$\begin{aligned} Pch(t) &\leq Battery_{max} - E[t-1] \\ Pch(t) &\leq Pch_{tot}(t) \end{aligned}$$

The surplus energy from the PV panel $Pch(t)$ at time t is less or equal to the decision variable $Battery_{max}$, which is the stored energy, reduced by the previous state of the battery so that the battery is not overcharged. The $Pch(t)$ can be less than $Pch_{tot}(t)$ as there can be more energy available from the PV panel than storage in the battery.

Equation 14. The balance between available energy, charging and discharging.

$$PV\ kWh(t) - load\ kWh(t) = Pch_{tot}(t) + Pdis(t)$$

The $PV\ kWh(t)$ is the available energy at time t where the $load\ kWh(t)$ (where the peak load for the modelling period is adjusted to be 1 kW) is deducted to show the available energy at time t . The sum of $Pch_{tot}(t)$ and $Pdis(t)$

Equation 15. Total modelled energy

$$PV\ kWh(t) = PV_{size} \bullet PVpower(t)$$

The $PV\ kWh(t)$ is the optimal size of the PV panel where the decision variable PV size is multiplied with the 1 kW PV panel system.

Equation 16. Charging and discharging battery – binary variable with big M constant

$$Pch_{tot}(t) \geq 0$$

$$Pch_{tot}(t) - M(1 - binary(t)) \leq 0$$

$$Pdis(t) \leq 0$$

$$Pdis(t) + M(1 - binary(t)) \geq 0$$

$$PV\ kWh(t) - load\ kWh(t) \leq M(1 - binary(t))$$

A binary variable is introduced to ensure that the system is either charged or discharged. A constant M is defined large enough to always be larger than the variables $Pch_{tot}(t)$ and $Pdis(t)$ at all times t . It should not be oversized, as that makes the optimisation run slowly, and for this system it is set to 100. This means that if there is a surplus of energy from the PV panel then $binary(t)$ has to be 0. And therefore $Pdis(t)$ has to be 0 to not violate the less than or equal to 0 constraints. Similarly, the opposite will be valid when there is a deficit of energy from the $PV\ kWh(t)$.

References

- [1] UN, Transforming Our World: the 2030 Agenda for Sustainable Development | Department of Economic and Social Affairs [WWW Document], 2022. <https://sdgs.un.org/2030agenda>. (Accessed 21 September 2022) (A/RES/70/1).
- [2] F. Fuso Nerini, J. Tomei, L.S. To, I. Bisaga, P. Parikh, M. Black, A. Borrión, C. Spataru, V. Castán Broto, G. Anandarajah, B. Milligan, Y. Mulugetta, Mapping synergies and trade-offs between energy and the sustainable development goals, *Nat. Energy* 3 (2018) 10–15, <https://doi.org/10.1038/s41560-017-0036-5>.
- [3] IEA, IRENA, UNSD, World Bank, WHO, Tracking SDG 7: the Energy Progress Report, World Bank, Washington DC, 2021.
- [4] V. Aryanpur, B. O'Gallachoir, H. Dai, W. Chen, J. Glynn, A review of spatial resolution and regionalisation in national-scale energy systems optimisation models, *Energy Strategy Rev.* 37 (2021), 100702, <https://doi.org/10.1016/j.esr.2021.100702>.
- [5] S. Pfenninger, A. Hawkes, J. Keirstead, Energy systems modeling for twenty-first century energy challenges, *Renew. Sustain. Energy Rev.* 33 (2014) 74–86, <https://doi.org/10.1016/j.rser.2014.02.003>.
- [6] IEA ETSAP, IEA-ETSAP | Times [WWW Document]. TIMES, 2022. <https://iea-etsap.org/index.php/etsap-tools/model-generators/times>. (Accessed 24 May 2022).
- [7] F. Riva, H. Ahlberg, E. Hartvigsson, S. Pachauri, E. Colombo, Electricity access and rural development: review of complex socio-economic dynamics and causal diagrams for more appropriate energy modelling, *Energy Sustain. Dev.* 43 (2018) 203–223, <https://doi.org/10.1016/j.esd.2018.02.003>.
- [8] N. Narayan, Z. Qin, J. Popovic-Gerber, J.-C. Diehl, P. Bauer, M. Zeman, Stochastic load profile construction for the multi-tier framework for household electricity access using off-grid DC appliances, *Energy Effic* 13 (2020) 197–215, <https://doi.org/10.1007/s12053-018-9725-6>.
- [9] S. Few, J. Barton, P. Sandwell, R. Mori, P. Kulkarni, M. Thomson, J. Nelson, C. Candelise, Electricity demand in populations gaining access: Impact of rurality and climatic conditions, and implications for microgrid design, *Energy Sustain. Dev.* 66 (2022) 151–164, <https://doi.org/10.1016/j.esd.2021.11.008>.
- [10] P. Sandwell, N.L.A. Chan, S. Foster, D. Nagpal, C.J.M. Emmott, C. Candelise, S. J. Buckle, N. Ekins-Daukes, A. Gambhir, J. Nelson, Off-grid solar photovoltaic systems for rural electrification and emissions mitigation in India, *Sol. Energy Mater. Sol. Cells*, Life cycle, environmental, ecology and impact analysis of solar technology 156 (2016) 147–156, <https://doi.org/10.1016/j.solmat.2016.04.030>.
- [11] N. Stevanato, F. Lombardi, G. Guidicini, L. Rinaldi, S.L. Balderrama, M. Pavicević, S. Quoilin, E. Colombo, Long-term sizing of rural microgrids: accounting for load evolution through multi-step investment plan and stochastic optimization, *Energy Sustain. Dev.* 58 (2020) 16–29, <https://doi.org/10.1016/j.esd.2020.07.002>.
- [12] F.A. Plazas-Niño, N.R. Ortiz-Pimiento, E.G. Montes-Páez, National energy system optimization modelling for decarbonization pathways analysis: a systematic literature review, *Renew. Sustain. Energy Rev.* 162 (2022), 112406, <https://doi.org/10.1016/j.rser.2022.112406>.
- [13] M.I. Yliruka, S. Moret, F. Jalil-Vega, A.D. Hawkes, N. Shah, The trade-off between spatial resolution and uncertainty in energy system modelling, in: Y. Yamashita, M. Kano (Eds.), *Computer Aided Chemical Engineering*, 14 International Symposium on Process Systems Engineering, Elsevier, 2022, pp. 2035–2040, <https://doi.org/10.1016/B978-0-323-85159-6.50339-0>.
- [14] Bradley, Magnanti Hax, *Applied Mathematical Programming* [WWW Document], 1977. <https://web.mit.edu/15.053/www/AMP-Chapter-09.pdf>. (Accessed 31 January 2023).
- [15] P. Ciller, S. Lumberras, Electricity for all: the contribution of large-scale planning tools to the energy-access problem, *Renew. Sustain. Energy Rev.* 120 (2020), 109624, <https://doi.org/10.1016/j.rser.2019.109624>.
- [16] F. Kemausuor, E. Adkins, I. Adu-Poku, A. Brew-Hammond, V. Modi, Electrification planning using Network Planner tool: the case of Ghana, *Energy Sustain. Dev.* 19 (2014) 92–101, <https://doi.org/10.1016/j.esd.2013.12.009>.
- [17] A. Korkovelos, B. Khavari, A. Sahlberg, M. Howells, C. Arderne, The role of open access data in geospatial electrification planning and the achievement of SDG7. An OnSSET-based case study for Malawi, *Energies* 12 (2019) 1395, <https://doi.org/10.3390/en12071395>.
- [18] D. Mentis, M. Howells, H. Rogner, A. Korkovelos, C. Arderne, E. Zepeda, S. Siyal, C. Taliotis, M. Bazilian, A. de Roo, Y. Tanvez, A. Oudalov, E. Scholtz, Lighting the World: the first application of an open source, spatial electrification tool (OnSSET)

- on Sub-Saharan Africa, *Environ. Res. Lett.* 12 (2017), 085003, <https://doi.org/10.1088/1748-9326/aa7b29>.
- [19] M. Zeyringer, S. Pachauri, E. Schmid, J. Schmidt, E. Worrell, U.B. Morawetz, Analyzing grid extension and stand-alone photovoltaic systems for the cost-effective electrification of Kenya, *Energy Sustain. Dev.* 25 (2015) 75–86, <https://doi.org/10.1016/j.esd.2015.01.003>.
- [20] M.I. Howells, T. Alfstad, D.G. Victor, G. Goldstein, U. Remme, A model of household energy services in a low-income rural African village, *Energy Pol.* 33 (2005) 1833–1851, <https://doi.org/10.1016/j.enpol.2004.02.019>.
- [21] N. Moksnes, A. Korkovelos, D. Mentis, M. Howells, Electrification pathways for Kenya—linking spatial electrification analysis and medium to long term energy planning, *Environ. Res. Lett.* 12 (2017), 095008, <https://doi.org/10.1088/1748-9326/aa7e18>.
- [22] I. Pappis, A. Sahlberg, T. Walle, O. Broad, E. Eludoyin, M. Howells, W. Usher, Influence of electrification pathways in the electricity sector of Ethiopia—policy implications linking spatial electrification analysis and medium to long-term energy planning, *Energies* 14 (2021) 1209, <https://doi.org/10.3390/en14041209>.
- [23] J.-P. Carvallo, B.J. Shaw, N.I. Avila, D.M. Kammen, Sustainable low-carbon expansion for the power sector of an emerging economy: the case of Kenya, *Environ. Sci. Technol.* 51 (2017) 10232–10242, <https://doi.org/10.1021/acs.est.7b00345>.
- [24] M.V. Rocco, E. Fumagalli, C. Vigone, A. Miserocchi, E. Colombo, Enhancing energy models with geo-spatial data for the analysis of future electrification pathways: the case of Tanzania, *Energy Strategy Rev.* 34 (2021), 100614, <https://doi.org/10.1016/j.esr.2020.100614>.
- [25] P.A. Trotter, N.J. Cooper, P.R. Wilson, A multi-criteria, long-term energy planning optimisation model with integrated on-grid and off-grid electrification – the case of Uganda, *Appl. Energy* 243 (2019) 288–312, <https://doi.org/10.1016/j.apenergy.2019.03.178>.
- [26] T. Brown, D. Schlachtberger, A. Kies, S. Schramm, M. Greiner, Synergies of sector coupling and transmission reinforcement in a cost-optimised, highly renewable European energy system, *Energy* 160 (2018) 720–739, <https://doi.org/10.1016/j.energy.2018.06.222>.
- [27] S. Pfenninger, B. Pickering, Calliope: a multi-scale energy systems modelling framework, *J. Open Source Softw.* 3 (2018) 825, <https://doi.org/10.21105/joss.00825>.
- [28] M. Groissböck, Are open source energy system optimization tools mature enough for serious use? *Renew. Sustain. Energy Rev.* 102 (2019) 234–248, <https://doi.org/10.1016/j.rser.2018.11.020>.
- [29] World Bank, Access to Electricity (% of Population) | Data [WWW Document], 2021, <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS>. (Accessed 8 March 2021).
- [30] African Development Bank (AfDB) Group, COUNTRY PRIORITY PLAN AND DIAGNOSTIC OF THE ELECTRICITY SECTOR, African Development Bank (AfDB) Group, 2021.
- [31] M.D. Wilkinson, M. Dumontier, I.J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, L.B. da Silva Santos, P.E. Bourne, J. Bouwman, A. J. Brookes, T. Clark, M. Crosas, I. Dillo, O. Dumon, S. Edmunds, C.T. Evelo, R. Finkers, A. Gonzalez-Beltran, A.J.G. Gray, P. Groth, C. Goble, J.S. Grethe, J. Heringa, P.A.C. 't Hoen, R. Hooft, T. Kuhn, R. Kok, J. Kok, S.J. Lusher, M. E. Martone, A. Mons, A.L. Packer, B. Persson, P. Rocca-Serra, M. Roos, R. van Schaik, S.-A. Sansone, E. Schultes, T. Sengstag, T. Slater, G. Strawn, M.A. Swertz, M. Thompson, J. van der Lei, E. van Mulligen, J. Velterop, A. Waagmeester, P. Wittenburg, K. Wolstencroft, J. Zhao, B. Mons, The FAIR Guiding Principles for scientific data management and stewardship, *Sci. Data* 3 (2016), 160018, <https://doi.org/10.1038/sdata.2016.18>.
- [32] M. Howells, H. Rogner, N. Strachan, C. Heaps, H. Huntington, S. Kypros, A. Hughes, S. Silveira, J. DeCarolis, M. Bazillian, A. Roehrl, OSeMOSYS: the open source energy modeling system, *Energy Pol.* 39 (2011) 5850–5870, <https://doi.org/10.1016/j.enpol.2011.06.033>.
- [33] F. Gardumi, A. Shivakumar, R. Morrison, C. Taliotis, O. Broad, A. Beltramo, V. Balderan, M. Howells, J. Hörsch, T. Niet, Y. Almulla, E. Ramos, T. Burandt, G. P. Sidderrama, G.N. Pinto de Moura, E. Zepeda, T. Alfstad, From the development of an open-source energy modelling tool to its application and the creation of communities of practice: the example of OSeMOSYS, *Energy Strategy Rev.* 20 (2018) 209–228, <https://doi.org/10.1016/j.esr.2018.03.005>.
- [34] Mikul Bhatia, Niki Angelou, Beyond Connections : Energy Access Redefined (TECHNICAL REPORT No. 008/15), Working Paper, 2015.
- [35] G. Falchetta, S. Pachauri, S. Parkinson, E. Byers, A high-resolution gridded dataset to assess electrification in sub-Saharan Africa, *Sci. Data* 6 (2019), <https://doi.org/10.1038/s41597-019-0122-6>.
- [36] C.D. Elvidge, M. Zhizhin, T. Ghosh, F.-C. Hsu, J. Taneja, Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019, *Rem. Sens.* 13 (2021) 922, <https://doi.org/10.3390/rs13050922>.
- [37] Facebook, Kenya: High Resolution Population Density Maps - Humanitarian Data Exchange [WWW Document], 2021, <https://data.humdata.org/dataset/high-resolution-population-density-maps-ken>. (Accessed 27 September 2021).
- [38] KPLC, Kenya - Kenya Electricity Network - ENERGYDATA.INFO [WWW Document], 2020, <https://energydata.info/dataset/kenya-kenya-electricity-net-work>. (Accessed 8 June 2021).
- [39] A.R. Jordan, P. Balla, L. Njagi, S. Banerjee, Kenya - Roads - ENERGYDATA.INFO [WWW Document], 2021, <https://energydata.info/dataset/kenya-roads-1>. (Accessed 8 June 2021).
- [40] A.R. Jordan, P. Balla, L. Njagi, S. Banerjee, Kenya - Overview of Off-Grid Electricity Service Areas - ENERGYDATA.INFO [WWW Document], 2020, <https://energydata.info/dataset/kenya-overview-of-off-grid-electricity-service-areas>. (Accessed 8 June 2021).
- [41] KPLC, Kenya - Primary Substations - ENERGYDATA.INFO [WWW Document], 2020, <https://energydata.info/dataset/kenya-primary-substations>. (Accessed 8 June 2021).
- [42] KPLC, Kenya - Distribution Transformers - ENERGYDATA.INFO [WWW Document], 2017, <https://energydata.info/dataset/kenya-distribution-transformers>. (Accessed 8 June 2021).
- [43] M. Kumm, M. Taka, J.H.A. Guillaume, Data from: Gridded Global Datasets for Gross Domestic Product and Human Development Index over 1990–2015, 2019, <https://doi.org/10.5061/DRYAD.DK1J0>.
- [44] I. Staffell, S. Pfenninger, Using bias-corrected reanalysis to simulate current and future wind power output, *Energy* 114 (2016) 1224–1239, <https://doi.org/10.1016/j.energy.2016.08.068>.
- [45] S. Pfenninger, I. Staffell, Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data, *Energy* 114 (2016) 1251–1265, <https://doi.org/10.1016/j.energy.2016.08.060>.
- [46] P.J. Burke, Z. Csereklyei, Understanding the energy-GDP elasticity: a sectoral approach, *Energy Econ.* 58 (2016) 199–210, <https://doi.org/10.1016/j.eneco.2016.07.004>.
- [47] E.W. Dijkstra, A note on two problems in connexion with graphs, *Numer. Math.* (1959), 269–271.
- [48] Pathfinder, Facebook Research, 2019.
- [49] C. Arderne, C. Zorn, C. Nicolas, E.E. Koks, Predictive mapping of the global power system using open data, *Sci. Data* 7 (2020) 19, <https://doi.org/10.1038/s41597-019-0347-4>.
- [50] B.J. van Ruijven, J. Schers, D.P. van Vuuren, Model-based scenarios for rural electrification in developing countries, *Energy* 38 (2012) 386–397, <https://doi.org/10.1016/j.energy.2011.11.037>.
- [51] Global Modeling and Assimilation Office (GMAO), MERRA-2, Goddard Space Flight Center Distributed Active Archive Center (GSFC DAAC), Greenbelt, MD, USA, 2015, <https://doi.org/10.5067/JVAFLP11CSIV>.
- [52] M.Z. Jacobson, V. Jadhav, World estimates of PV optimal tilt angles and ratios of sunlight incident upon tilted and tracked PV panels relative to horizontal panels, *Sol. Energy* 169 (2018) 55–66, <https://doi.org/10.1016/j.solener.2018.04.030>.
- [53] N. Bolson, P. Prieto, T. Patzek, Capacity factors for electrical power generation from renewable and nonrenewable sources, *Proc. Natl. Acad. Sci. USA* 119 (2022), e2205429119, <https://doi.org/10.1073/pnas.2205429119>.
- [54] S.P. Koko, Optimal battery sizing for a grid-tied solar photovoltaic system supplying a residential load: a case study under South African solar irradiance, *Energy Rep.* ICPE 2021 - The 2nd International Conference on Power Engineering 8 (2022) 410–418, <https://doi.org/10.1016/j.egyr.2022.02.183>.
- [55] AirSquid, Answer to “Python Pulp linear optimisation for off-grid PV and battery system”, Stack Overflow (2022).
- [56] R. Loulou, G. Goldstein, A. Kanudia, A. Lettila, U. Remme, Documentation for the TIMES Model, 2016.
- [57] Energy and Petroleum Regulatory Authority (EPRA) Kenya, Geothermal Development Company, KenGen, Kenya Power, Rural Electrification and Renewable Energy Corporation, KETRACO, Nuclear Power and Energy Agency, 2021. Updated least cost power development plan Study period 2020–2040.
- [58] Energy and Petroleum Regulatory Authority (EPRA) Kenya, Geothermal Development Company, KenGen, Kenya Power, Rural Electrification and Renewable Energy Corporation, KETRACO, Nuclear Power and Energy Agency, LEAST COST POWER DEVELOPMENT PLAN 2021–2030, 2021.
- [59] C.D. Elvidge, M. Zhizhin, T. Ghosh, F.-C. Hsu, J. Taneja, Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019, *Rem. Sens.* 13 (2021) 922, <https://doi.org/10.3390/rs13050922>.
- [60] IEA, IEA Sankey Diagram [WWW Document], 2019, <https://www.iea.org/sankey/#?c=Kenya&s=Final%20consumption>. (Accessed 22 September 2022).
- [61] N.J. Williams, P. Jaramillo, K. Campbell, B. Musanga, I. Lyons-Galante, Electricity consumption and load profile segmentation analysis for rural micro grid customers in Tanzania, in: 2018 IEEE PES/IAS PowerAfrica, IEEE, Cape Town, 2018, pp. 360–365, <https://doi.org/10.1109/PowerAfrica.2018.8521099>.
- [62] KPLC, Half-hourly Load Profile 2015, 2015.
- [63] IEA, Electricity Transmission and Distribution - ETSAP [WWW Document], 2014, https://iea-etsap.org/E-TechDS/PDF/E12_el-t&d_KV_Apr2014_GSOK.pdf. (Accessed 18 October 2021).
- [64] I. Pappis, Strategic Low-Cost Energy Investment Opportunities and Challenges towards Achieving Universal Access (SDG7) in Each African Nation, 2021, <https://doi.org/10.5281/zenodo.5810601>.
- [65] NREL (National Renewable Energy Laboratory), Annual Technology Baseline, National Renewable Energy Laboratory, Golden, CO, 2021.
- [66] International Labour Organization, Global Wage Report 2020–21. Wages and Minimum Wages in the Time of COVID-19, ILO, Geneva, 2020 (No. ISBN 978-92-2-031945-1).
- [67] D.T. Hermann, N. Donatien, T.K.F. Armel, T. René, A feasibility study of an on-grid PV/Wind/Battery/Diesel for residential buildings under various climates in Cameroon, *Energy Technol.* 9 (2021), 2100615, <https://doi.org/10.1002/ente.202100615>.
- [68] C. Mokhtara, B. Negrou, A. Bouferrouk, Y. Yao, N. Settou, M. Ramadan, Integrated supply-demand energy management for optimal design of off-grid hybrid renewable energy systems for residential electrification in arid climates, *Energy Convers. Manag.* 221 (2020), 113192, <https://doi.org/10.1016/j.enconman.2020.113192>.
- [69] A. Korkovelos, B. Khavari, A. Sahlberg, M. Howells, C. Arderne, The role of open access data in geospatial electrification planning and the achievement of SDG7. An

- OnSSET-based case study for Malawi, *Energies* 12 (2019) 1395, <https://doi.org/10.3390/en12071395>.
- [70] N. Moksnes, J. Rozenberg, O. Broad, C. Taliotis, M. Howells, H. Rogner, Determinants of energy futures—a scenario discovery method applied to cost and carbon emission futures for South American electricity infrastructure, *Environ. Res. Commun.* 1 (2019), 025001, <https://doi.org/10.1088/2515-7620/ab06de>.
- [71] Kitui County, *Kitui County Integrated Development Plan 2018-2022*, 2018.
- [72] Narok County, *Narok County Integrated Development Plan 2018-2022*, 2018.
- [73] World Bank, Access to Electricity (% of Population) - Kenya | Data [WWW Document], 2021. <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=KE>. (Accessed 18 October 2021).
- [74] Ministry of Energy Kenya, *Updated Least Cost Power Development Plan 2017-2037* [WWW Document], 2018. <http://gak.co.ke/wp-content/uploads/2019/02/Updated-Least-Cost-Power-Development-Plan-2017-2022-min.pdf>. (Accessed 7 April 2021).
- [75] B. Khavari, A. Sahlberg, W. Usher, A. Korkovelos, F. Fuso Nerini, The effects of population aggregation in geospatial electrification planning, *Energy Strategy Rev.* 38 (2021), 100752, <https://doi.org/10.1016/j.esr.2021.100752>.
- [76] GADM, GADM [WWW Document], 2021. https://gadm.org/download_country_v3.html. (Accessed 27 February 2021).
- [77] NREL (National Renewable Energy Laboratory), Useful Life [WWW Document], 2021. <https://www.nrel.gov/analysis/tech-footprint.html>. (Accessed 18 February 2021).
- [78] D.M. Gioutsos, K. Blok, L. van Velzen, S. Moorman, Cost-optimal electricity systems with increasing renewable energy penetration for islands across the globe, *Appl. Energy* 226 (2018) 437–449, <https://doi.org/10.1016/j.apenergy.2018.05.108>.
- [79] D. Yamegueu, Y. Azoumah, X. Py, N. Zongo, Experimental study of electricity generation by Solar PV/diesel hybrid systems without battery storage for off-grid areas, *Renew. Energy* 36 (2011) 1780–1787, <https://doi.org/10.1016/j.renene.2010.11.011>.
- [80] N. Moksnes, *UN Sustainable Development Goals from a Climate Land Energy and Water Perspective for Kenya (MSc)*, KTH, Stockholm, 2016.
- [81] A.Q. Jakhrani, A.-K. Othman, A.R.H. Rigit, S.R. Samo, Estimation of carbon footprints from diesel generator emissions, in: 2012 International Conference on Green and Ubiquitous Technology, IEEE, Jakarta, 2012, pp. 78–81, <https://doi.org/10.1109/GUT.2012.6344193>.
- [82] IEA, *World energy outlook 2019*, vol. 810, OECD Publ, 2019, <https://doi.org/10.1787/caf32f3b-en>.
- [83] World Bank, Population, Total | Data [WWW Document], 2020. <https://data.worldbank.org/indicator/SP.POP.TOTL>. (Accessed 14 October 2021).